



Discrete element modelling of a chevron patterned conveyor belt and a transfer chute

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ABSTRACT

The Discrete Element Method (DEM) is increasingly used to analyse the design of conveyor systems. In this study, the capability of, and accuracy with which DEM can predict the flow of a non-cohesive granular material through a transfer chute was investigated. The model included a Chevron patterned belt, an impact plate, hood, and rock box. The results were compared to measurements using high-speed footage. The model parameter values were determined and the particle shape modelled as single spheres and multi-spheres. To decrease computation time, particle up-scaling, simplification of the shape model, and decreasing the contact stiffness were investigated. Results showed that DEM could accurately predict the flow characteristics, and was more accurate than analytical approaches. It is concluded that DEM can be used with confidence in modelling similar applications if a calibrated parameter set is used, and the guidelines presented here in terms of particle scaling, are adhered to.

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1. Introduction

In the fields of agriculture, food processing, pharmaceuticals and mining, equipment such as screw conveyors, belt conveyors, bucket elevator and bucket wheels play an essential role in the conveying of granular materials in bulk form. Belt conveyors are usually preferred due to its versatility and the ability to transport large volumes of material over long distances. However, due to plant layout and changes in elevation and direction, transfer points are needed [1]. The main purpose of a transfer point is to receive the material from the feeding belt and to guide it onto the receiving belt. The design should consider aspects such as central and symmetric loading on the receiving conveyor, minimising chute and belt impact angles (reducing particle attrition, dust emissions and structural wear), minimising the differential velocity between the material and the receiving conveyor at the point of loading, and at the same time prevent spillage and blockages [2]. It is challenging to design ideal transfer points based on rule-of-thumb principles, scale model prototyping or using analytical models. Although analytical models consider material properties such as friction, bulk density and cohesion [3–6], they are generally limited to a 2-D lumped parameter model and neglect the inter-particle interaction and particle shape.

The Discrete Element Method (DEM) has developed into a popular design tool and provides visualisation of the material flow, and also enables the user to model complex interactions between discrete particles and particles-and-walls (structures), where the contact forces and moments, particle velocities, rotations and displacements are calculated at

each timestep [7]. However, before any DEM model can be used as an accurate predictive tool, an appropriate set of calibrated parameter values should be specified. These parameters include the particle density, sliding and rolling coefficients of friction, coefficient of restitution (damping), and the particle shape and size distribution [8]. According to Coetzee [9], there are two approaches for selecting the DEM parameters, namely, the *direct measuring approach* and the *bulk calibration approach*. In the *direct measuring approach*, the material properties at the particle and/or contact level are directly measured. However, if assumptions and simplifications are made in terms of particle shape and size and contact physics, the accuracy of any prediction in bulk behaviour remains uncertain if the micro- and/or meso-properties are directly measured and implemented.

On the other hand, in the *bulk calibration approach*, laboratory scale tests such as measuring the angle of repose are conducted. These tests are then replicated in DEM and the parameter values adjusted to fit the measured bulk (macro) behaviour [10]. The philosophy is that this *calibration* process accounts for all assumptions and simplifications. Depending on the contact model(s) used and the number of parameters to calibrate, an appropriate number of independent experiments should be used to find a unique set of parameter values [11].

2. Analytical models for bulk flow

Various analytical models exist to predict the flow of bulk material. These models are based on semi-empirical and analytical methods to predict, for example, the trajectory as the material leaves the feeding conveyor [12–16], the impact forces on a plate (wall) [12], and the flow through a transfer hood [3,17,18]. Analytical models have shown

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to be reasonably accurate in predicting the material flow onto an impact plate, but failed in other areas [19]. In this study, a combination of trajectory models [4–6,17,20], Korzen's impact plate model [12] and Roberts's hood model [3,18] was used.

3. DEM modelling of conveyor transfer points

Various studies have modelled and analysed conveyor transfer chutes using DEM. One of the earliest demonstrations of this was by Nordell in 1994 [21], and one of the first implementations of DEM as a predictive design tool was by Hustulid in 1996 and 1998 [22,23]. For researchers and practitioners to have sufficient confidence in the ability of DEM to accurately model and predict the flow of material through a transfer point, the models should be thoroughly validated. Below follows a summary of previous studies where DEM (a) was used in demonstration examples, showcasing its capabilities without any validation, (b) was compared to analytical models but not to experiments or observations, (c) was used in industrial scale case studies and qualitatively validated, and (d) was quantitatively validated by comparing the results to laboratory scale measurements and observations.

3.1. Example demonstrations

Example demonstrations, such as those presented by [24–28], are important to show the general capabilities of a method, but cannot provide any quantitative validation. Katterfeld et al. [29] demonstrated the combined use of DEM and Computational Fluid Dynamics (CFD) to model and predict dust emissions from a transfer chute. Dewicki and Mustoe [30] described some of the typical problems associated with belt conveyor transfer points and demonstrated how DEM can be used to address it. Doroszuk and Krol [31] analysed a rock box and a hood and spoon type transfer chute. The material trajectory was compared to analytical solutions, and although this was based on a case study, no comparison with measurements or observations were made. Gutierrez and Garate [32] analysed a transfer chute using different feeding rates, belt inclinations, the absence and presence of cohesion in the material, and a transfer angle of 0 and 90 in 16 unique cases. The general flow of the material was analysed, but no verification or validation of the model was offered. Grima [25] modelled the flow of cohesive copper ore through a rock box type chute. The DEM parameters were calibrated in large scale laboratory experiments, but the model of the chute was not validated. However, the effects of particle shape was identified, and it was concluded that **care should be taken when spherical particles are used to model cohesive materials, as it can result in undesirable flow behaviour**.

3.2. Comparisons with analytical models

Ilic et al. [33] analysed a hood and spoon type transfer chute, and compared the results to analytical models, however, no experimental validation was performed. Ilic et al. [34,35] compared the velocity profile through a hood and spoon to analytical results, with mixed success (see Section 3.3). Hastie and Wypych [15,36,37] compared the material trajectory and flow through a hood and spoon to analytical results (see Section 3.4).

3.3. Industry case studies

Industry case studies allow for qualitative visual comparisons to be made, but in most cases still lack quantitative comparisons [10,38]. Industry partners are unlikely to stop production facilities to be instrumented, and normal working conditions make it difficult to safely measure and observe all aspects of the material flow.

Grima et al. [39] investigated the flow of **wet and sticky** (cohesive) ore through a troublesome transfer chute. The parameter values we

obtained through laboratory scale calibration tests, and then used to model the existing chute. Qualitative comparisons with field observations were used to verify the model.

In a series of papers, Ilic et al. modelled transfer chutes characterised by accelerated flow [34] and shearing flow [35]. In the first case [34], a **bifurcated chute** for transporting of coal was analysed. The DEM results of the velocity profile through the loading spoon differed significantly from that of an analytical model. Unfortunately, the results could not be quantitatively compared to measurements (verification was limited to the flow observed through access hatches), and the accuracy of both approaches remain unclear. In the second case [35], a stacker transfer chute for transporting of coal was analysed. The velocity profile through the hood was analysed in DEM and compared to that of an analytical model. Although both models predicted the velocity to be in the same range, the trends differed significantly. The results could not be compared to measurements or observations, and again the accuracy of both models remain unclear (see Section 8 for such a comparison).

Ilic [40] presented a case study where DEM was used to optimise an existing transfer chute. Although the model could not be validated, it was concluded that the chute and belt service life could be increased by approximately 132% and 139%, respectively.

Donohue et al. [41] used DEM to successfully improve Roy Hill's transfer chutes after it experienced blockages and adverse loading conditions. The DEM model was first qualitatively calibrated by modelling the existing scenario, before any design changes were considered. Where the facility experienced downtimes ranging from 7 to 30 h per week, it experienced downtimes of only 0–5 h per week after the improvements were implemented.

The case study by Groger and Katterfeld [42] is the most successful published study in terms of quantitative validation. A large scale industrial transfer station was instrumented to measure the force on an impact plate (wall), the force on three idler sets directly below the impact zone of the receiving belt, and the mass flow rate was measured by a belt scale. It was concluded by the authors that the model accurately predicted the force on the impact wall (error <17%), and to a lesser degree, the force on the idler sets. In the current paper, an error ≤10% is considered to be accurate.

3.4. Laboratory scale comparisons

Laboratory scale experiments provide the best opportunity to thoroughly validate the DEM models. Material flow rates can be accurately monitored and controlled, forces on impact plates can be measured using load cells, and the flow can be recorded using high-speed photography. Below follows a short summary of previous studies, including the shortcomings of each which are addressed in this paper.

Chen et al. [43] modelled the flow of material through a stacker chute, and compared the results to scaled laboratory tests. However, no quantitative comparisons were performed, but qualitatively, good results were reported.

Hastie and Wypych [15,36,37] modelled the flow trajectory of non-cohesive materials as it discharged from the head pulley, and also the flow of the material through a transfer hood and spoon. The results were compared to measurements and analytical models. However, they did not analyse the flow against an impact plate, into rock box, or through a chute. The effect of particle scale was also not investigated.

Grima et al. [10,44] modelled the flow of non-cohesive material against an impact plate, and investigated the effects of particle shape and scale. They, however, obtained different results in terms of particle scale and shape as to what was obtained here (see Section 7.6 and Section 7.7 respectively). The flow through a transfer hood, rock box, and chute was not investigated.

Carr et al. [45–48] modelled the build-up of cohesive materials on a vertical and inclined plate. The mass and shape of the build-up material was compared to measurements and observations. However, no other aspects of a transfer chute was analysed.

Ilic [49] modelled the belt loading profile (surcharge angle) of various cohesive and non-cohesive materials using laboratory and site tests. This included the pressure exerted by the material on the belt, which was measured using TekScan technology. The material behaviour in the belt transition zone, and the trajectory during discharge was modelled and compared to measurements and observations. However, the flow through the remainder of the transfer chute was not analysed.

4. Objectives and contribution

The purpose of this study was to investigate and validate the accuracy of DEM in predicting the flow of non-cohesive granular material through a laboratory scale transfer point, including an impact plate, a hood, a rock box, and a chute. The DEM simulations were validated by comparing the results to that of analytical models and high-speed footage of the material flow, which was analysed using particle image velocimetry (PIV). Where possible, the input parameter values were directly measured by implementing various material tests (*direct measuring approach*). The remaining parameter values were either calibrated based on the *bulk calibration approach*, or taken from literature. Where values were taken from literature, a sensitivity study was undertaken to quantify the model response to each parameter. Single spheres (referred to as *balls*) and multi-sphere particles (referred to as *clumps*) were used to investigate the effects of particle shape. Furthermore, the maximum particle scale factor, which still accurately predicted the material flow onto an impact plate, into and out of a rock box and chute blockage and build-up, was determined.

None of the previous and published studies (Section 3.4) have analysed an impact plate, a hood, a rock box and a chute in a single

attempt. The advantage of this is that the same experimental set-up, test material and conditions, and DEM model were used to analyse all of these different components, thus directly comparing the accuracy of DEM in modelling the different flow scenarios. This is the first study to model the material build-up in a conveyor transfer chute, including the effects of particle scale, and also the first study where DEM was used to model a chevron patterned belt (all previous studies used flat/smooth belts).

5. Material characterisation

Corn grains were used for all tests because of its irregular particle shape and size, and ease to experiment with, Fig. 1. The DEM software, PFC by Itasca [50], was used with the linear spring-dashpot (LSD) contact model [7]. A summary of the final parameter values is given in Table 2 with reference to the sections or literature where it was obtained.

5.1. Measured properties

The parameters which could be determined using the *direct measuring approach* are presented in this section.

5.1.1. Particle shape and size

Due to corn grains having several shapes and sizes, four grains were randomly selected and scanned with a 3D laser scanner, Fig. 1a. The resulting Stereolithography (STL) models were used to generate three sets of clumps, consisting of 3, 5 and 10 constituent spheres respectively. ASG3D software [51] was used to generate these clumps in an optimised manner. The 3-, 5- and 10-clump models are shown in

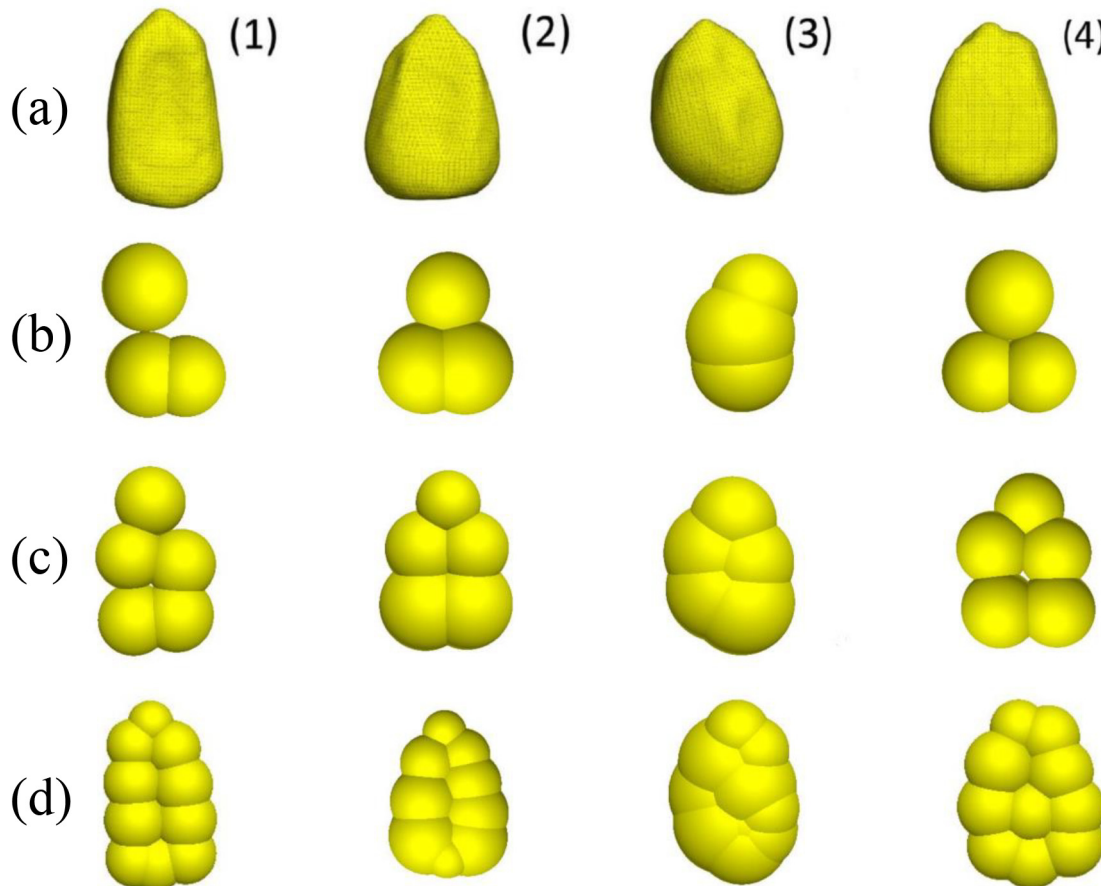


Fig. 1. (a) Four scanned corn grains [8], (b) optimised 3-clump, (c) 5-clump and (d) 10-clump models.

Fig. 1b, Fig. 1c and Fig. 1d, respectively. In this study, the 3-clump particles were mostly used, and the 5- and 10-clump particles were only used to quantify the sensitivity of the model to the particle shape. The scanned STL models were used to obtain the volume of the grains, from which the volume equivalent spherical diameter was calculated to range between $D_{ev} = 7.5$ mm and 9.6 mm. Balls (single spheres) were also used, however, rolling resistance was added to these particles to account for the effects of particle shape. The ball particles were created having a diameter equal to the equivalent volume diameter.

5.1.2. Contact damping

The coefficients of restitution (COR) between corn grains and aluminium and polycarbonate (materials in the transfer chute) were determined by drop tests, where the velocity directly before and after impact was recorded using particle image velocimetry software, Tracker [52]. The guidelines proposed by Hastie [53] were followed, which state that only the tests where the particles rebounded vertically should be considered (otherwise the impact angle and rotation of the particles before and after impact should also be accounted for). A minimum of 10 successful drops were considered in this study, and based on the COR, the particle-wall critical damping ratio ζ_{pw} was calculated using the relation (Itasca, 2019),

$$COR = \exp \frac{-\zeta_{pw} \pi}{\sqrt{1-\zeta_{pw}^2}} \quad (1)$$

The results showed no significant difference for the two surfaces, Table 1. Wang et al. [54] measured the COR between corn grains and a zincified plate to be between 0.635 and 0.613, which correlates very well with our results. It was further assumed that the particle-particle damping μ_{pp} is equal to the particle-wall damping [55].

5.1.3. Particle-wall friction

Four strategies were implemented to measure the particle-wall sliding friction μ_{pw} between corn grains and aluminium, polycarbonate and rubber. The four strategies can be divided into two groups, with two different experimental setups. The first setup was an inclined tester, depicted in Fig. 2a. Individual grains (strategy 1) and a single layer of grains glued to a wooden plate (strategy 2) were placed on the plane, which was then tilted until the sample started to slide. These strategies provided only the coefficient of friction under relatively low contact forces.

The experimental setups for strategies 3 and 4 are shown in Fig. 2b and Fig. 2c respectively, and was used to measure the coefficient of friction under higher contact forces. A rotating platform, covered with the material of interest, was used. With strategy 3, the wooden plate samples were placed on the platform, and with strategy 4, loose grains were poured into a diameter 70 mm aluminium cylinder, approximately two particles high, and placed on the platform. The platform was then rotated while the shear force on the sample was measured using a S2M 50 N HBM load cell. A normal load was applied to the wooden plate and to an aluminium lid placed inside the cylinder, using known weights up to a maximum of 2 kg. To reduce friction between the aluminium cylinder and the surface material in strategy 4, a Teflon ring (2 mm thick) was glued underneath the cylinder, Fig. 2c. The shear force (≈ 0.14 N) due to only the Teflon surface and the empty cylinder was measured and considered negligible. The shear

velocity could be set between $10.5 \text{ mm} \cdot \text{s}^{-1}$ and $52.4 \text{ mm} \cdot \text{s}^{-1}$, which had no significant effect on the results.

The inclined tester with single grains (strategy 1) resulted in a mean coefficient of friction $\mu_{pw} = 0.34$ (standard deviation 0.06) with Aluminium. The wooden plate samples, also on the inclined tester (strategy 2), resulted in $\mu_{pw} = 0.29$ (0.05). On the rotating platform, the wooden plate samples (strategy 3), resulted in a slightly lower value $\mu_{pw} = 0.20$ (0.04), and the filled cylinder (strategy 4) resulted in $\mu_{pw} = 0.24$ (0.05).

Only the wooden plate samples on the inclined tester were used to measure the friction with the polycarbonate and rubber surfaces. For the polycarbonate surface, a friction value of $\mu_{pw} = 0.26$ (0.05) was measured, and plane angles greater than 45 were found with the rubber surface and as a result, the coefficient of friction for rubber surfaces was selected as 1.

To conclude, a significant variation in friction was found between the four strategies; therefore these results were merely used as a reference point and further studies were conducted to quantify the sensitivity of the transfer chute model to the friction parameters (see Section 7.5 and Section 9.1). This emphasises the difficulty in accurately measuring the material properties at the micro- or meso- scale if the particles are relatively small and irregular in shape, one of the disadvantages of the *direct measuring approach* as detailed in [9].

5.2. Calibration of the particle density

The bulk density was determined by measuring the mass and fill height of loosely poured grains in a glass cylinder, resulting in an average bulk density of $\rho_b = 824.73 \text{ kg} \cdot \text{m}^{-3}$. The particle density was determined by the water displacement method, similar to the approach used by [56,57]. A sample of grains was weighed, then submerged under water to determine the total volume, resulting in an average particle density of $\rho_p = 1249.95 \text{ kg} \cdot \text{m}^{-3}$. Similar values were obtained by Coetzee [8] and González-Montellano et al. [58].

The particle density was calibrated in DEM by modelling the cylindrical container, as used in the experiment, and filling it with 3-clump particles. Once settled under gravity, the particle density was iteratively adjusted until the bulk density was within 1% of the measured value. The calibrated particle density was $\rho_p = 1366.63 \text{ kg} \cdot \text{m}^{-3}$ which was slightly higher than the measured particle density $\rho_p = 1249.95 \text{ kg} \cdot \text{m}^{-3}$. This 9% difference can be attributed to the simplified shape, resulting in a porosity slightly different from the real material. Nonetheless, the bulk density was always accurately modelled. Scaling the particle size (and the container) by a factor of 8 had no significant effect on the calibrated particle density, and the value of $\rho_p = 1366.63 \text{ kg} \cdot \text{m}^{-3}$ was used in all subsequent models.

5.3. Parameters from literature

This study used the same corn grains as calibrated by Coetzee [8], where a combination of test results were used to identify a unique (or close to unique) parameter set for each particle shape. Based on these findings, the particle-particle sliding and rolling coefficients of friction of the balls were initially set to $\mu_{pp} = 0.20$ and $\mu_{rpp} = 0.125$, respectively. For the 3-clumps, the initial values of $\mu_{pp} = 0.15$ and $\mu_{rpp} = 0$ were used. Additional friction parameter combinations were also investigated in various sections as indicated in Table 2.

The particle-particle contact stiffness was also calibrated by Coetzee [8] using a confined uniaxial compression test. It was also shown that the calibrated stiffness of $k_{pp} = 100 \text{ kN} \cdot \text{m}^{-1}$ could be scaled down by a factor 10 in ring shear tests without any adverse effects, while in draw down tests it could be scaled down by a factor of 100. Thus, in this study, where the material was under relatively low confining stresses, the particle-particle contact stiffness was set to $k_{pp} = 1 \text{ kN} \cdot \text{m}^{-1}$. The particle-wall stiffness was always set to $k_{pw} = 2 k_{pp}$, i.e. $k_{pw} = 2 \text{ kN} \cdot \text{m}^{-1}$. The effect of contact stiffness was, however, further investigated in Section 7.4 for the impact force on an impact plate.

Table 1
Measured contact damping in drop tests.

Interface	COR	ζ_{pw}	Std Dev
	Mean	Mean	
Corn - Aluminium	0.64	0.14	0.03
Corn - Polycarbonate	0.66	0.13	0.04

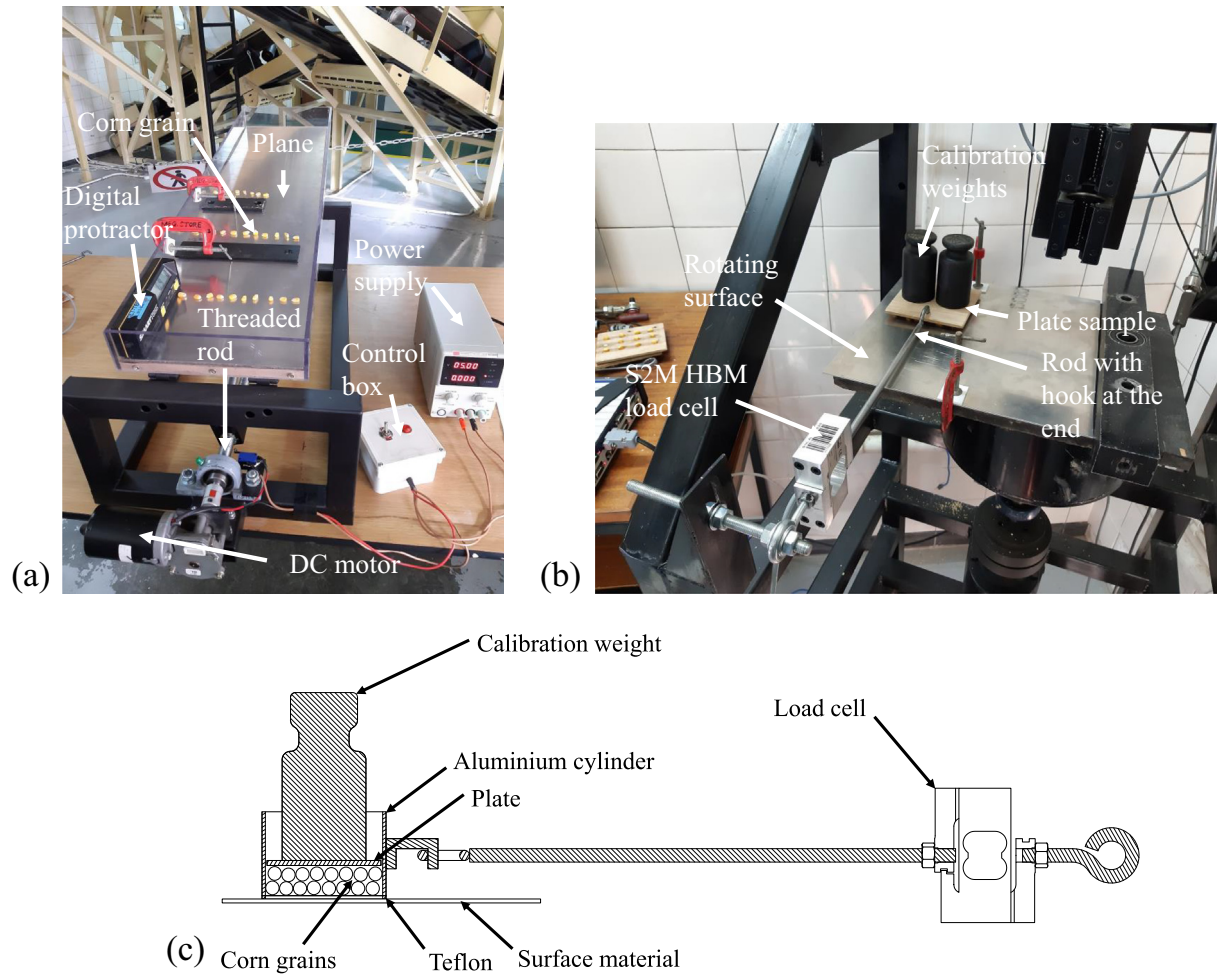


Fig. 2. Particle-wall friction test setups of the (a) inclined tester (strategies 1 and 2) and rotating surface test with the (b) wooden plate samples (strategy 3) and (c) cylindrical container (strategy 4).

Table 2
Summary of DEM parameter values for corn grains.

Symbol	Description	Section(s)	Range/Value	Unit
ρ_b	Bulk density (loosely poured)	5.2	824.73	$\text{kg} \cdot \text{m}^{-3}$
ρ_p	Particle density (water displacement)	5.2	1366.63	$\text{kg} \cdot \text{m}^{-3}$
D_{ev}	Equivalent volume diameter upper limit	5.1.1	9.60	mm
	Equivalent volume diameter lower limit	5.1.1	7.50	mm
k_{pw}	Stiffness particle-aluminium	5.3 and 7.4	$2^a - 200$	$\text{kN} \cdot \text{m}^{-1}$
	Stiffness particle-polycarbonate	5.3	2	$\text{kN} \cdot \text{m}^{-1}$
	Stiffness particle-rubber	5.3	1	$\text{kN} \cdot \text{m}^{-1}$
k_{pp}	Stiffness particle-particle	5.3 and 7.4	$1^a - 100$	$\text{kN} \cdot \text{m}^{-1}$
ζ_{pw}	Damping particle-wall	5.1.2 and 7.3	$0.05 - 0.14^a - 1.00$	$\text{N} \cdot \text{s} \cdot \text{m}^{-1}$
ζ_{pp}	Damping particle-particle	5.1.2, 7.3 and 9.1	$0.05 - 0.14^a - 0.50$	$\text{N} \cdot \text{s} \cdot \text{m}^{-1}$
μ_{pw}	Sliding friction particle-aluminium	5.1.3	$0.20 - 0.25^a - 0.90$	–
	Sliding friction particle-polycarbonate	5.1.3 and 8	$0.20 - 0.26^a - 0.35$	–
	Sliding friction particle-rubber	5.1.3	1	–
μ_{rpw}	Rolling friction particle-rubber (balls)	7.7.2	$0.00 - 0.10^a - 0.90$	–
μ_{pp}	Sliding friction particle-particle (clumps)	5.3, 7.5 and 9.1	$0.05 - 0.15^a - 0.40$	–
	Sliding friction particle-particle (balls)	5.3, 7.7.2 and 9.3	$0.15 - 0.25^a - 0.70$	–
μ_{rpp}	Rolling friction particle-particle (balls)	5.3, 7.7.2 and 9.3	$0.125 - 0.50^a - 0.80$	–

^a Parameter value that delivered the most accurate results.

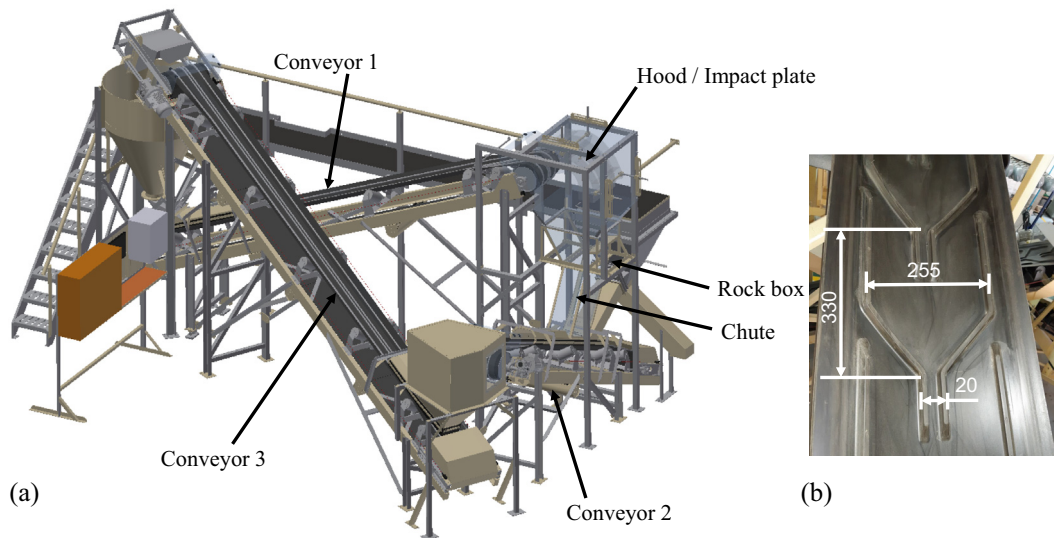


Fig. 3. (a) Belt conveyor test facility (CAD model) and (b) chevron pattern on inclined belts (dimensions in mm).

6. Experiments and DEM models

The experimental setups and related DEM models are presented.

6.1. Belt conveyor test facility

The belt conveyor test facility is situated in the Granular Materials Research Group (GMRG) laboratory in the Department of Mechanical and Mechatronic Engineering, University of Stellenbosch, as shown in Fig. 3a. The facility is also equipped with high-speed cameras (Olympus i-Speed 3) which were strategically placed to record the material flow. The setup with three conveyors ensures continuous circulation of the material at a constant flow rate as set by a control valve. In this study, the transfer point between conveyors 1 and 2 was investigated. This included the feeding conveyor 1 (6 m in length), an impact plate or hood placed at the point of discharge and in the path of the material flow, a rock box, and a chute feeding onto the receiving conveyor 2.

6.2. Conveyor belt

Since the conveyors were inclined at 23° , a chevron patterned belt was used to prevent the material from sliding and rolling backwards and to provide more control over the mass flow rate, especially at higher belt speeds. The chevron pattern had a V-shape (Fig. 3b) with the distance between successive patterns 330 mm and the “V” having a

maximum width of 255 mm, and the bottom opening a width of 20 mm. The total belt width was 450 mm with a trough angle of 35° .

The belt had a troughed profile along its length, shaped by equally spaced idler rollers and flattened out in the transition zone towards the head pulley, Fig. 4. The shape of the flat part of the belt (without the patterns) was accurately measured and a STL model created in CAD [59]. In the DEM model, each triangular facet of the flat belt was assigned the velocity of the conveyor. Around the head pulley, the velocity vector of each facet was calculated using the facet's centroid as reference position and the pulley radius. These facets (flat belt and head pulley) did not actually displace, but the set velocity was used as a conveyor boundary condition when resolving particle-facet contacts.

However, the same approach could not be followed with the chevron patterns, which had to displace to create the desired effects. The patterns were modelled in CAD and sliced into 10 (rigid) pieces in the direction of travel, Fig. 4. The pieces were all assigned an initial velocity vector in the direction of belt motion. However, to ensure that the patterns closely followed the belt profile in the transition zone, the pieces were each individually slaved to a facet. At each time step, the flat belt facet closest to each pattern piece was found, and the facet's boundary velocity vector assigned to the piece, i.e. updating the piece's velocity vector. Furthermore, in the transition zone, the pieces were all given a constant rotation (around an axis in the direction of belt travel) to ensure that it stayed perpendicular to the belt, Fig. 4. When the pieces reached the head pulley, they were assigned a constant angular velocity to rotate around the pulley.

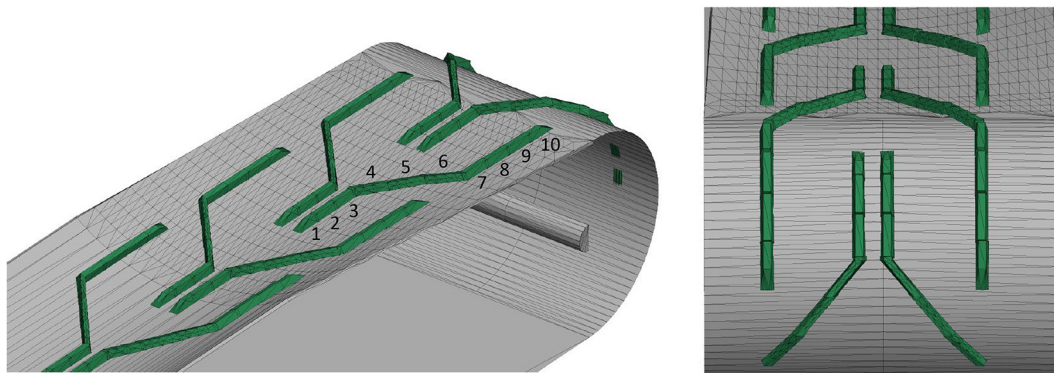


Fig. 4. Chevron patterned conveyor belt in DEM.

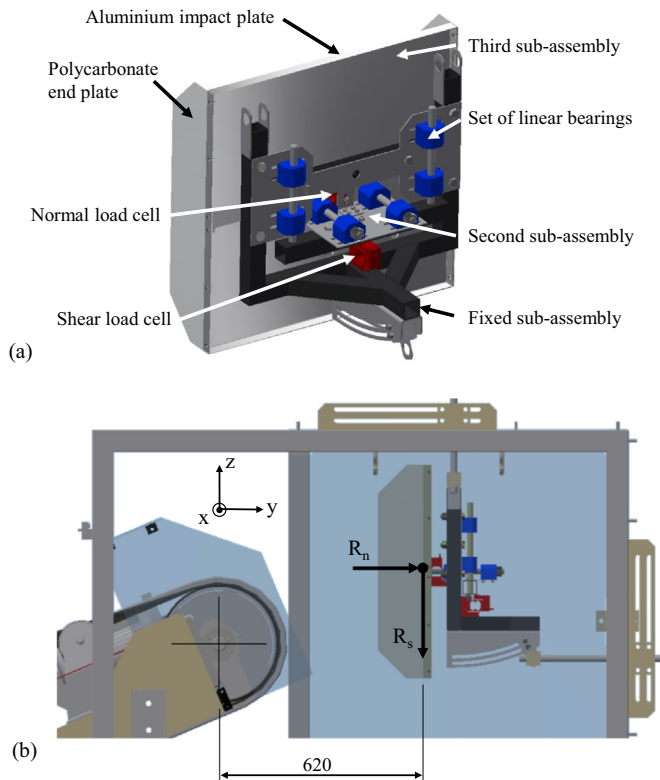


Fig. 5. (a) Impact plate design and (b) relative position (dimensions in mm).

6.3. Impact plate

The impact plate consisted of three sub-assemblies: the one fixed to the transfer chute support structure, the second free to move only tangentially relative to the fixed sub-assembly, and the third free to move only in the direction normal to the second sub-assembly, Fig. 5a. The fixed sub-assembly was manufactured from square tubing, and the second sub-assembly connected to it using a set of linear bearings, thus allowing relative motion only in one direction (measuring the shear/tangential force on the impact plate). The third sub-assembly attached to the second sub-assembly using an identical set of linear bearings, allowing relative motion between the two sub-assemblies only in one direction (measuring the normal force on the impact plate). Although the two sets of linear bearings allowed for relative motion, it was restricted by fitting two load cells, the first (S2M 200 N HBM) between the fixed and first sub-assemblies for measuring the shear force, and the second load cell (S2M 500 N HBM) between the second and third

sub-assemblies for measuring the normal impact force. The impact plate itself was fixed to the third sub-assembly and manufactured from aluminium with two polycarbonate end plates to guide the flow but still allow for observation. The load cell data was recorded using a HBM Quantum MX840B data acquisition system with Catman Easy [60] software and the sampling frequency was set to 500 Hz.

The DEM model is shown in Fig. 5b, indicating the position of the impact plate relative to the feeding conveyor's head pulley. The normal force R_n and the shear force R_s were calculated based on the contact data and sampled at a frequency of 500 Hz.

6.4. Transfer hood

The purpose of a transfer hood is to change the direction of the material flow and reduce the horizontal velocity component, without significantly decelerating the material [31]. A transfer hood is usually located in a hood-spoon configuration or before an intermediate chute. It is important to recognise the hood and spoon as two distinct sections having different design approaches [61]. However, for designing an optimum transfer chute, the two sections should be assessed together. In this study, only the transfer hood was evaluated.

Fig. 6a shows the DEM model with the distance from the centre of the head pulley to the centre of the hood's curvature (note that the centre of the hood was 17 mm below the head pulley's centre, which is not indicated in the Figure). The hood consisted of polycarbonate panels which were bolted to two polycarbonate side panels to ensure that the material did not leave the hood sideways. The curved part consisted of flat panels, arranged on a radius of $R = 373$ mm. The last panel at the outlet made an angle of -107° to the y-axis (counter-clockwise taken as positive as indicated in Fig. 6a). This ensured that the material was directed down the centre of the chute.

6.5. Rock box

By successfully incorporating rock boxes in transfer chutes, a major reduction in impact wear can be achieved. However, the location and arrangement of the rock box are fundamental to comply with the desired function of a transfer chute, i.e. to keep the mass flow rate constant and centrally load the material on the receiving conveyor [61].

The simulated rock box geometry and location in the transfer chute are shown in Fig. 7. The rock box had a 136 mm ledge, angled at 15° and a $10\text{ mm} \times 10\text{ mm}$ lip was mounted at the edge to ensure a layer of particles was entrapped.

6.6. Chute

Transfer chutes usually clog up due to poor design and control of the conveyor (feeding and receiving) speeds. An effective transfer chute

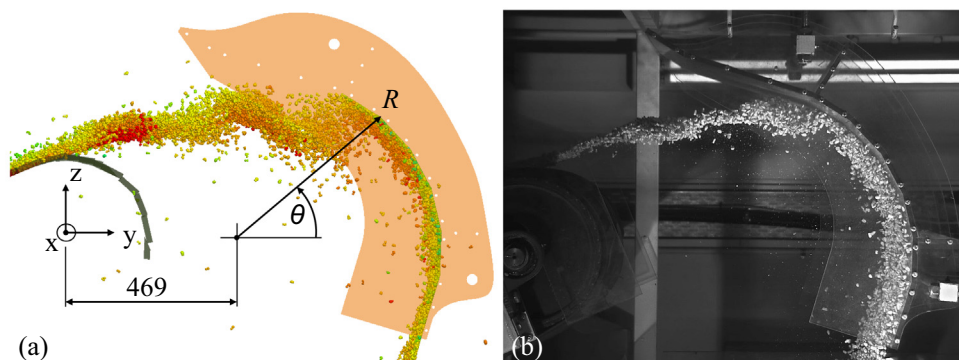


Fig. 6. Transfer hood material flow from (a) DEM with 3-clump particles and (b) high-speed footage ($V_b = 4\text{ m/s}$ and $m = 8\text{ kg/s}$).

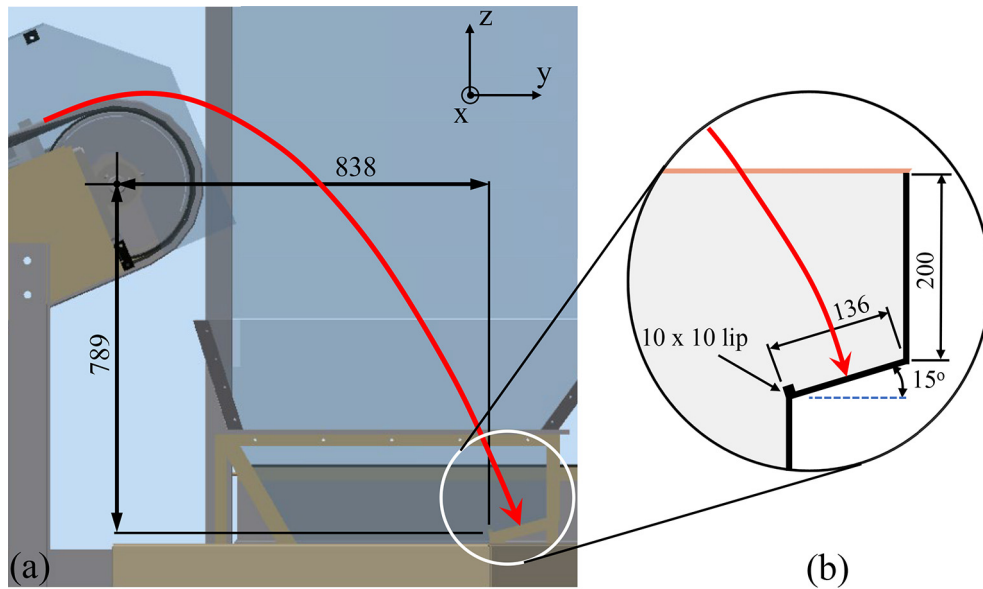


Fig. 7. Rock box (a) relative position and (b) size, dimensions in mm (depth of 500 mm).

minimises the differential velocity between the material and the receiving conveyor, while also minimising wear in the chute [62].

The chute was located between conveyor 1 and 2, Figure 3a, having an inlet opening of 480 mm × 396 mm, outlet of 80 mm × 396 mm and a total height of 845 mm, Fig. 8a. Note that the outlet was set to 80 mm only for this test, in all previous tests the outlet was 300 mm.

7. Results: vertical impact plate

The impact plate analysis was conducted for belt speeds of $V_b = 3 \text{ m} \cdot \text{s}^{-1}$ and $4 \text{ m} \cdot \text{s}^{-1}$ with the mass flow rate ranging from $\dot{m} = 4 \text{ kg} \cdot \text{s}^{-1}$ to $12 \text{ kg} \cdot \text{s}^{-1}$. In all cases, except where the effects of particle shape was investigated, the 3-clump particles were used.

7.1. Comparing impact forces

The time signal of the normal and shear force components predicted by DEM are compared to the measurements in Fig. 9. The 'gap' in the experimental data at approximately 3 s (Fig. 9a) was caused by the splice

in the conveyor belt where the chevron pattern did not join perfectly, resulting in a larger gap between subsequent patterns.

A Fast Fourier Transforms (FFT) was performed on the normal force-time signals and the major frequency was found to be 9.17 Hz, 12.22 Hz and 15.25 Hz for belt speeds of $3 \text{ m} \cdot \text{s}^{-1}$, $4 \text{ m} \cdot \text{s}^{-1}$ and $5 \text{ m} \cdot \text{s}^{-1}$ respectively. This compared well to the theoretically calculated frequencies based on the measured distance between patterns, namely 9.09 Hz, 12.12 Hz and 15.15 Hz respectively (difference less than 1%). The FFT was also performed on the DEM results, and the major frequency was found to be 9.00 Hz, 12.00 Hz and 15.00 Hz respectively (1% error compared to theoretical frequencies). This indicates that the DEM model accurately predicted the force frequency content caused by the chevron pattern, which could be important for design purposes to avoid structural resonance frequencies, for example.

DEM predicted peak forces in the normal direction of approximately 70 N (Fig. 9b) and approximately 10 N in the shear direction (Fig. 9d). These peak forces were significantly larger than the measured forces of approximately 50 N (Fig. 9a) and 5 N (Fig. 9c) respectively. One reason for this is that the physical impact plate and supporting structures were elastic, and the small deflections (vibration) and inertia might influence the forces measured by the load cells. In DEM, the plate is

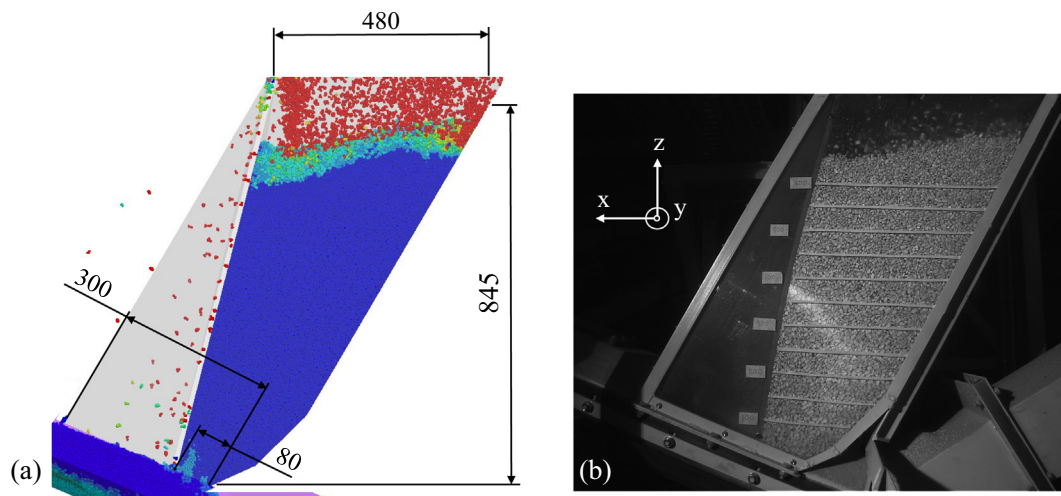


Fig. 8. (a) Chute dimensions and DEM model of build-up (3-clump particles), (b) the equivalent experimental setup (dimensions in mm and the depth of the chute was 396 mm).

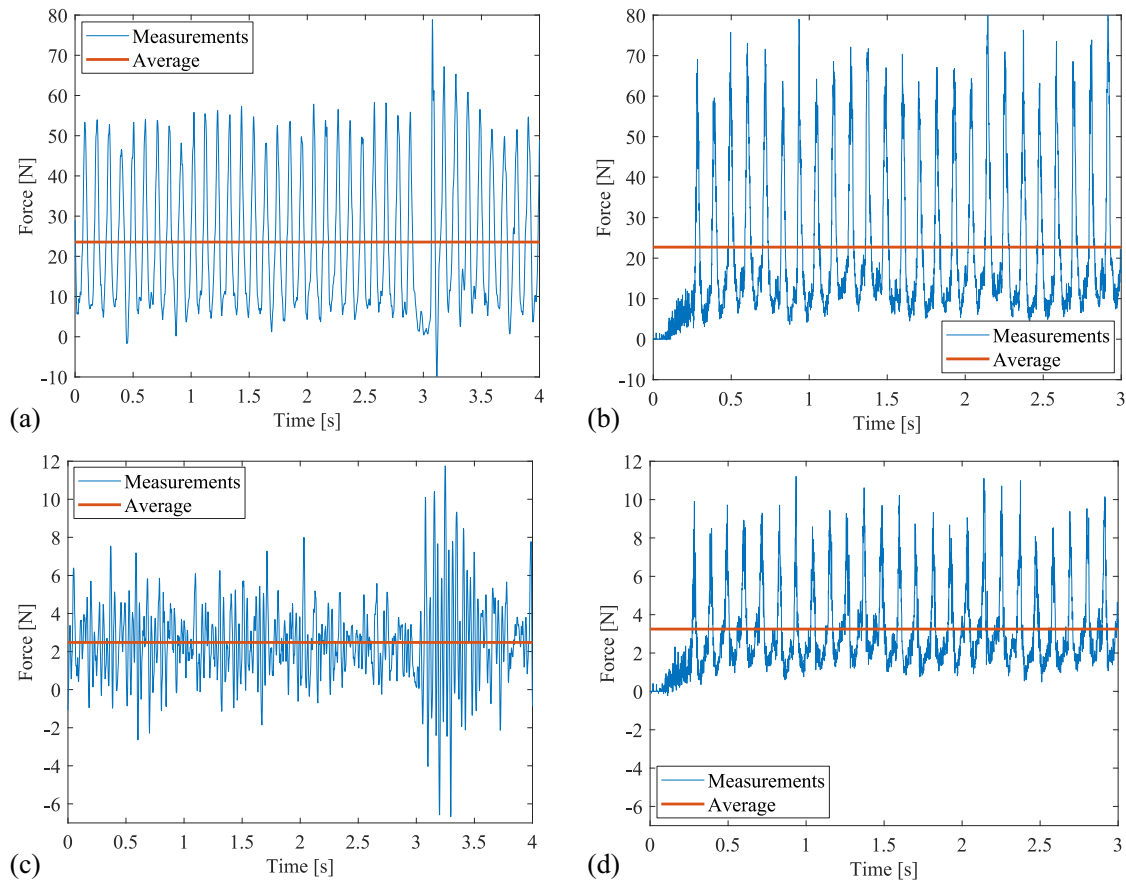


Fig. 9. Normal force time signal from the (a) experimental tests and (b) DEM, and shear force time signal from the (c) experimental tests and (d) DEM ($V_b = 3$ m/s and $\dot{m} = 8$ kg/s).

considered perfectly rigid and the predicted forces are purely due to the particle-plate collisions.

Two abrasive wear mechanisms are involved in these applications [63,64], impact wear (Finnie's model [65]) that occurs when the particle makes contact with the plate (or any other structure), and sliding wear (Archard's model [66]) when the particle slides along the plate surface. It is believed that the impact wear will be influenced by the peak forces but needs further investigation. However, further comparisons were made by taking the time-averaged forces in the normal direction R_n and shear direction R_s as summarised in Table 3 for the experiments, the analytical model [12], and two DEM models. In the first DEM model, the chevron pattern was included, while in the second model, a flat belt was modelled. It is clear from the results that in all four cases the normal force increased with an increase in belt speed and mass flow rate. The force increased linearly with an increase in mass flow rate (see Fig. 11). The results further showed that the patterned DEM model accurately predicted the normal force with the error

ranging between 1.8% and 9.0% (under-predicting in all, but one case). This is in line with the results by Grima [10] who found the difference to range between 0.3% and 29.4%. However, in Grima's case, the DEM model over-predicted the results in all but one case (varying the belt speed and mass flow rate).

The flat belt model, however, failed to accurately predict the normal force, under-predicting it by approximately 10% to 25%. A detailed analysis showed that with the patterned belt, the material velocity at the conveyor discharge was the same as the belt velocity, while that of the flat belt was approximately 0.89 times slower (the mass flow rate was controlled to be the same in both models). Thus, it is concluded that the effect of the chevron pattern should not be neglected for the simulation of patterned conveyor belts, and it was included in all subsequent models. The shear forces were relatively small, and although both DEM models accurately predicted the trends, they failed to accurately predict the magnitude of the shear force, over-predicting it in all cases.

Table 3

Normal R_n and shear R_s forces on the impact plate from the patterned DEM, flat belt DEM, and analytical models.

Belt speed [m·s ⁻¹]	Mass flow [kg·s ⁻¹]	Experiment		Patterned DEM model			Flat belt DEM model			Analytical model		
		R_n	R_s	R_n	R_s	Error*	R_n	R_s	Error*	R_n	R_s	Error*
3	4	11.73	0.61	11.97	1.74	2.0 %	10.15	1.81	-10.5 %	10.34	0.08	-11.8 %
	8	23.54	2.48	22.71	3.25	-3.5 %	19.00	3.28	-19.3 %	20.69	0.15	-12.1 %
	12	35.89	4.37	32.73	4.64	-8.8 %	27.00	4.48	-24.8 %	31.03	0.23	-13.5 %
4	4	15.57	0.39	14.17	1.45	-9.0 %	12.43	1.77	-20.2 %	14.72	3.32	-5.5 %
	8	28.13	0.99	27.59	3.41	-1.9 %	23.01	3.57	-18.2 %	29.45	6.65	4.7 %
	12	41.03	2.33	40.29	5.13	-1.8 %	32.63	4.90	-20.5 %	44.17	9.97	7.7 %

*Error % calculated with respect to measured normal force.

Values in red (> 10 %) are considered inaccurate.

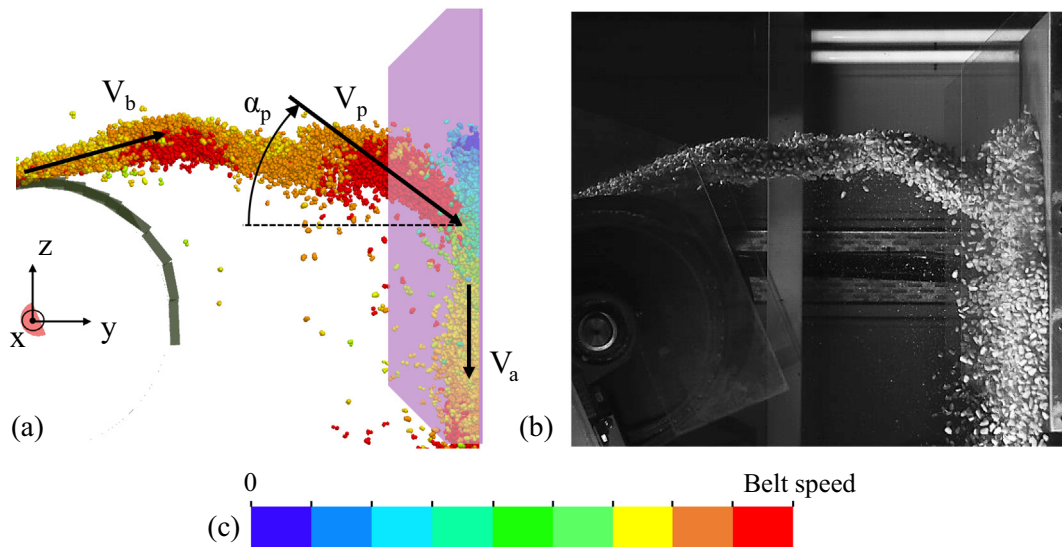


Fig. 10. Vertical impact plate material flow from (a) DEM with 3-clump particles and (b) high-speed footage ($V_b = 3 \text{ m/s}$ and $\dot{m} = 12 \text{ kg/s}$), (c) with a colour bar indicating the particles' velocity magnitude.

The analytical model showed acceptable accuracy in predicting the normal force at a belt speed of $4 \text{ m} \cdot \text{s}^{-1}$ with errors $< 8\%$. However, for a belt speed of $3 \text{ m} \cdot \text{s}^{-1}$ it under-predicted the normal force by approximately 12%. This result is in line with the findings by Grima [10] who found the Korzen model to be accurate within 23.7%, 14.3%, 2.5% for a belt speeds of $3 \text{ m} \cdot \text{s}^{-1}$, $4 \text{ m} \cdot \text{s}^{-1}$ and $5 \text{ m} \cdot \text{s}^{-1}$ respectively. The analytical model also failed to accurately predict the shear force in all cases, significantly under-predicting it for the lower belt speeds, and over-predicting for the higher belt speed. Grima [10] also found the model by Korzen to be inaccurate in predicting the shear force, over-predicting it in all cases investigated. This could be attributed to a limitation in the model, which assumes stationary flow (see [10,12] for details).

7.2. Comparing material flow characteristics

Fig. 10a and b show a comparison between DEM and a high-speed image of the material flow from the head pulley to the point where the material free-falls after impacting the plate (note the colour bar in Fig. 10c for an indication of the particles' velocity). In this zone, the material impact velocity V_p , impact angle α_p and outflow velocity V_a were investigated, as defined in Fig. 10a.

The impact velocity V_p from PIV and DEM was determined by implementing spatial and time averaging. Spatial averaging was performed by taking the average velocity of all particles across the flow stream (into the viewing plane, Fig. 10), just before the point of contact. Time averaging was performed by the taking the average of the spatially averaged velocities over the whole duration of steady flow (a period of 18.5 s). The material outflow velocity V_a was measured at a location 150 mm below the centre of the impact plate.

In the experiment there was no significant change in the impact velocity with a change in mass flow rate, only in belt speed (see Table 4).

The DEM results showed similar trends, and in general were in good agreement. The DEM model under-predicted the impact velocity in all cases, and was more accurate for a belt speed of $3 \text{ m} \cdot \text{s}^{-1}$ (error $< 3\%$), than for $V_b = 4 \text{ m} \cdot \text{s}^{-1}$ (error $< 5.3\%$). Grima [10] also found DEM to under-predict the impact velocity, and found larger errors ranging from 4.2% to 23.1%. The analytical model over-predicted the impact velocity in all cases, with the error $< 6.1\%$.

DEM accurately predicted the impact angle for a belt speed $V_b = 3 \text{ m} \cdot \text{s}^{-1}$ with the largest deviation only 2.25° . The analytical model over-predicted the angle by approximately 4.5° . For $V_b = 4 \text{ m} \cdot \text{s}^{-1}$, the measured angle for all mass flow rates was negative, indicating that the material was still moving upwards at impact. The DEM and analytical models, however, both predicted small positive angles. However, the angles were all relatively small, indicating that the flow was almost horizontal at impact, and the models were still accurate within 4° .

The DEM model failed to accurately predict the material outflow velocity V_a for a belt speed $V_b = 3 \text{ m} \cdot \text{s}^{-1}$ and mass flow rates of $8 \text{ kg} \cdot \text{s}^{-1}$ and $12 \text{ kg} \cdot \text{s}^{-1}$ (over-predicting with errors of 12.1% and 19.9% respectively). However, for a mass flow of $4 \text{ kg} \cdot \text{s}^{-1}$ the model was accurate with an error of 3.6%. For a belt speed of $V_b = 4 \text{ m} \cdot \text{s}^{-1}$ (and all mass flow rates), the model was accurate with the error $\leq 5.4\%$ (under-predicting in all cases). Grima [10] found DEM to accurately predict the outflow velocity with errors ranging from 3.9% to 5.6%. The analytical model under-predicted the outflow velocity in all cases, with errors ranging from 9.1% to 19.4%.

7.3. The influence of contact damping

The critical damping ratio between the corn grains and aluminium, ζ_{pw} , was determined as 0.14 (COR = 0.64) in Section 5.1.2, by implementing a drop test. It was assumed that the contact damping

Table 4
Measured and predicted impact velocity V_p , angle α_p and outflow velocity V_a on a vertical impact plate.

Belt speed [m·s ⁻¹]	Mass flow rate [kg·s ⁻¹]	Impact velocity, V _p [m·s ⁻¹]						Impact angle, α _p [°]			Outflow velocity, V _a [m·s ⁻¹]					
		Experiment	DEM		Analytical		Experiment	DEM	Analytical	Experiment	DEM		Analytical			
			V _p	Error*	V _p	Error*					V _a	Error*	V _a	Error*		
3	4	2.78	2.73	-1.80 %	2.94	5.76 %	15.57	16.75	19.89	2.20	2.28	3.64 %	1.78	-19.10 %		
	8	2.77	2.72	-1.81 %	2.94	6.14 %	15.53	16.84	19.89	2.06	2.31	12.14 %	1.78	-13.59 %		
	12	2.78	2.70	-2.88 %	2.94	5.76 %	15.18	17.43	19.89	1.96	2.35	19.90 %	1.78	-9.18 %		
4	4	3.59	3.43	-4.46 %	3.68	2.51 %	-4.88	0.72	1.02	2.27	2.20	-3.08 %	1.83	-19.38 %		
	8	3.58	3.39	-5.31 %	3.68	2.79 %	-3.70	1.57	1.02	2.22	2.10	-5.41 %	1.83	-17.57 %		
	12	3.58	3.46	-3.35 %	3.68	2.79 %	-4.31	2.09	1.02	2.24	2.20	-1.79 %	1.83	-18.30 %		

*Error % calculated with respect to the measured value.

Values in red (> 10 %) are considered inaccurate.

between the grains, ζ_{pp} , was the same as the particle-wall damping. Coetzee [55] showed that the damping ratio had no significant effect on the angle of repose with the largest change in angle 0.7° when the damping ratio was varied from 0.1 to 0.9 (COR ranging from 0.73 to 0.002 respectively).

The effect of varying ζ_{pw} and ζ_{pp} on the impact force and material flow was investigated here. It was found that the particle-wall damping ζ_{pw} had no significant effect on the impact force or material flow. This is in agreement with the results by Grima [10] who varied the damping ratio from $\zeta_{pw} = 0.03$ to $\zeta_{pw} = 0.33$ (coefficient of restitution from 0.90 to 0.33), and found the change in normal impact force to be $<1.9\%$.

Changing the particle-particle damping ζ_{pp} , however, had a significant effect. At a low value of $\zeta_{pp} = 0.05$, the impact force was underestimated and the effect was more pronounced for higher mass flow rates, Fig. 11a. However, for higher values $\zeta_{pp} \geq 0.14$, there was no significant effect. Qualitatively, for lower particle damping values, there was a large number of particles deviating from the mainstream leaving the head pulley, Fig. 11b. This was due to less energy dissipated during particle collisions. For a high ζ_{pp} , the impact position on the plate was slightly higher, and more particles splashed upwards after impact. It is concluded that the calibration of the contact damping is important when the material flow from a conveyor belt is to be accurately simulated, in the extreme case, the change in impact force was 8.6%.

7.4. The influence of contact stiffness

From the findings in Section 5.3, k_{pp} and k_{pw} were selected as $1 \text{ kN} \cdot \text{m}^{-1}$ and $2 \text{ kN} \cdot \text{m}^{-1}$, respectively. The effect of contact stiffness on the reaction force was investigated by repeating the simulations with $V_b = 3 \text{ m} \cdot \text{s}^{-1}$ and $\dot{m} = 8 \text{ kg} \cdot \text{s}^{-1}$ while scaling both stiffness' up by factors of 10 and 100, respectively. The results showed the reaction force to be insensitive to changes in the stiffness, with the maximum change in the normal and shear force 1.07% and 3.50%, respectively. Hence, the selected stiffness values were used in all subsequent models.

7.5. The influence of particle friction

Using the 3-clump particles, the particle sliding friction μ_{pp} was varied from 0.05 to 0.40 and the particle rolling friction μ_{rpp} from 0 to 0.80. The reaction forces, impact velocity and outflow velocity on the vertical impact plate were insensitive to these changes.

7.6. The influence of particle scale

The effect of the particle scale (3-clump particles) was investigated for a belt speed of $3 \text{ m} \cdot \text{s}^{-1}$ and mass flow rates between $4 \text{ kg} \cdot \text{s}^{-1}$ and $12 \text{ kg} \cdot \text{s}^{-1}$. Accurate predictions (error $\leq 6.49\%$) of the average

reaction force were obtained for scaling factors up to 9. However, only scaling factors up to 7 could accurately predict the force's frequency content. Scaling factors larger than 7, reduced the number of particles to such levels, that the frequency content was lost. Scaling factors up to 7 also had no significant effect on the material's discharge velocity, trajectory and the impact position on the plate.

Grima [10] scaled spherical particles up by a factor of 2 and 3, and showed that it had no significant effect on the average reaction force. However, it was shown that when the mass flow and belt velocity remained unchanged while the particles were scaled up, the number of particles in contact with the plate was reduced (reduction on contact resolution). When smaller particles were used, there was a smooth and even distribution of the contact forces over the plate, and with larger particles the contact forces were more concentrated in distinctive regions. This is important when the model is used to predict the wear on chute structures [40], in which case the scaling of particles should be carefully considered.

7.7. The influence of particle shape

7.7.1. Clumps

Up to this point, only the 3-clump particles were used. However, the DEM parameter values presented in Table 2 were also used for the 5-clump and 10-clump particles. The sliding friction μ_{pp} was set to 0.15 and the rolling friction μ_{rpp} to 0. The results showed that the particle shape had no significant effect on the impact velocity, impact angle or after-impact velocity.

In Section 7.1 (Fig. 9) it was reported that the 3-clump particles accurately predicted the average normal impact force (error $<9\%$), but over-predicted the peak values (error 36%). Changing the particle shape had no significant effect on the average impact force, but had an effect on the peak values, reducing the error to 8% and 9% for the 5-clump and 10-clump particles respectively. As a result, when the force peaks, and not only the average values, are of interest, more accurate particle shapes should be considered. This might become important when the model is used to predict material wear.

7.7.2. Balls

As discussed in Section 5.3, the sliding friction μ_{pp} and rolling friction μ_{rpp} for the balls were initially estimated as 0.20 and 0.125, respectively. A sensitivity study was performed to obtain the particle-wall rolling friction μ_{rpw} where the normal impact force (vector sum of the normal and shear components) was used as the calibration measure. It was found that for $\mu_{rpw} > 0.1$, it had no significant effect on the reaction forces, and as a result, the magnitude of the reaction forces depended solely on the inter-particle friction. Therefore, additional combinations of μ_{pp} and μ_{rpp} (with $\mu_{rpw} = 0.10$) were investigated, Table 5.

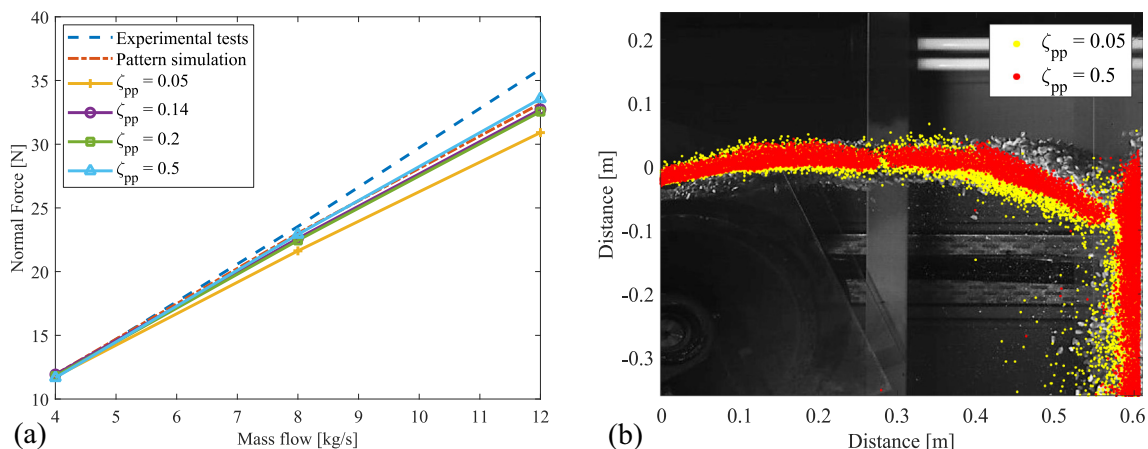


Fig. 11. Effect of ζ_{pp} on the (a) predicted average normal force and (b) material flow on a vertical plate ($V_b = 3 \text{ m/s}$).

Table 5Impact plate results using balls and various combinations of μ_{pp} and μ_{rp} with $\mu_{pw} = 0.10$, $V_b = 3 \text{ m/s}$ and $\dot{m} = 12 \text{ kg/s}$.

μ_{pp}	μ_{rp}	Reaction force and error			Impact velocity and error		
		R [N]	Compared to 3-clump model	Compared to experiment	$V_p [\text{m}\cdot\text{s}^{-1}]$	Compared to 3-clump model	Compared to experiment
0.15	0.18	29.72	-10.10 %	-17.81 %	2.60	-3.70 %	-6.37 %
0.20	0.125	30.42	-7.99 %	-15.87 %	2.64	-2.22 %	-4.93 %
0.20	0.80	32.11	-2.87 %	-11.20 %	2.71	0.37 %	-2.41 %
0.25	0.125	31.71	-4.08 %	-12.31 %	2.66	-1.48 %	-4.32 %
0.25	0.50	33.35	0.88 %	-7.77 %	2.76	2.22 %	-0.72 %
0.40	0.125	32.85	-0.64 %	-9.15 %	2.73	1.11 %	-1.69 %
0.70	0.125	33.67	1.85 %	-6.89 %	2.76	2.22 %	-0.61 %

μ_{pp}	μ_{rp}	Impact angle and error			After-impact velocity and error		
		$\alpha_p [^\circ]$	Compared to 3-clump model	Compared to experiment	$V_a [\text{m}\cdot\text{s}^{-1}]$	Compared to 3-clump model	Compared to experiment
0.15	0.18	18.62	6.83 %	22.66 %	2.36	0.43 %	20.41 %
0.20	0.125	17.90	2.70 %	17.92 %	2.38	1.28 %	21.43 %
0.20	0.80	16.16	-7.29 %	6.46 %	2.41	2.55 %	22.96 %
0.25	0.125	16.88	-3.16 %	11.20 %	2.39	1.70 %	21.94 %
0.25	0.50	15.08	-13.48 %	-0.66 %	2.43	3.40 %	23.98 %
0.40	0.125	15.98	-8.32 %	5.27 %	2.43	3.40 %	23.98 %
0.70	0.125	15.40	-11.65 %	1.45 %	2.42	2.98 %	23.47 %

Values in red (> 10 %) are considered inaccurate.

The results showed that the calibrated friction range by Coetzee [8] ($\mu_{pp} = 0.15$ to 0.25 and $\mu_{rp} = 0.125$ to 0.80) could not provide the necessary accuracy in predicting the reaction forces for all parameter combinations (error ranging from 12.3% to 17.8%). Setting $\mu_{pp} = 0.25$ and $\mu_{rp} = 0.50$ was the only combination within the calibrated range which accurately predicted the material impact force (7.77% error), impact velocity (0.72% error) and impact angle (0.66% error).

Setting μ_{pp} to larger values of 0.40 and 0.70 respectively, in combination with $\mu_{rp} = 0.125$, also provided accurate results with the maximum error of 9.15% in predicting the reaction force. The combinations $\mu_{pp} = 0.25$, $\mu_{rp} = 0.50$ and $\mu_{pp} = 0.70$, $\mu_{rp} = 0.125$ provided the most accurate results with errors in the same order.

No combination of parameters μ_{pp} and μ_{rp} could accurately predict the after-impact velocity V_a , and all predicted a significantly higher velocity (error ranging from 20% to 24%). Under the same flow conditions, the 3-clump particles were also unable to accurately predict the after-impact velocity with the error 20% (Table 4). This indicates that the error is not due to the particle shape, and should be further investigated.

Hastie and Wypych [15] found that the particle shape (spherical versus multi-spheres) had no significant effect on the material trajectory after it left the head pulley. However, they had to carefully calibrate the rolling friction for each shape independently, to ensure that the material reached the belt speed before the point of discharge.

Changing the shape from spherical to a multi-sphere particle (with two comprising spheres), Grima [10] found that the change in impact force was less than 3.1% and deemed insignificant. This is contrary to our results, where changes in the impact force was as large as 10% when the shape was changed from 3-clump to balls. Grima did not investigate the effects of particle shape on the outflow velocity.

8. Results: transfer hood

Only the 3-clump particles were considered for this analysis with belt speeds of $3 \text{ m}\cdot\text{s}^{-1}$ and $4 \text{ m}\cdot\text{s}^{-1}$ and mass flow rates of $4 \text{ kg}\cdot\text{s}^{-1}$,

$8 \text{ kg}\cdot\text{s}^{-1}$ and $12 \text{ kg}\cdot\text{s}^{-1}$. The position of impact and material velocity through the hood were compared to measurements, Fig. 6.

The analytical model used a combination of trajectory models [4–6,17,20], Korzen's impact plate model [12] and Roberts's hood model [3,18]. The trajectory models were used to determine the position of impact on the hood, the impact plate model was used to determine the material velocity just before and after impact and the hood model to determine the velocity profile through the hood. The results are presented with respect to the angular position θ along the hood's perimeter, Fig. 6a (with the outlet at $\theta = -30^\circ$).

As shown in Fig. 6 for $V_b = 4 \text{ m}\cdot\text{s}^{-1}$ and $\dot{m} = 8 \text{ kg}\cdot\text{s}^{-1}$, qualitatively the DEM results agreed well to the experiments for all belt speeds and mass flow rates. The impact position on the transfer hood is summarised in Table 6 for $\dot{m} = 8 \text{ kg}\cdot\text{s}^{-1}$. The measurements were taken from the centre of the head pulley to the centre of the material stream at the point of impact. The DEM results correlated very well with the experiments, however, the analytical model predicted points of impact lower down (smaller z-coordinate and larger y-coordinate).

Fig. 12a shows the measured and predicted velocity profile through the hood for $V_b = 3 \text{ m}\cdot\text{s}^{-1}$ and $\dot{m} = 12 \text{ kg}\cdot\text{s}^{-1}$ including the effects of μ_{pw} . In the experiment, PIV was used to track the particles and determine the velocity at the centre of the stream and as close to the wall as possible. The velocity was only determined against the wall for the DEM models. The velocity profile at the wall was very accurately predicted by DEM, using the calibrated sliding friction between corn and

Table 6

Position of impact on the hood measured from the centre of the head pulley.

Belt speed $V_b [\text{m}\cdot\text{s}^{-1}]$	Impact coordinates [mm]					
	DEM		Experiment		Analytical	
	Y	Z	Y	Z	Y	Z
3	830.9	74.7	832.2	72.0	852.7	18.6
4	660.0	307.9	663.6	305.2	712.8	281.9

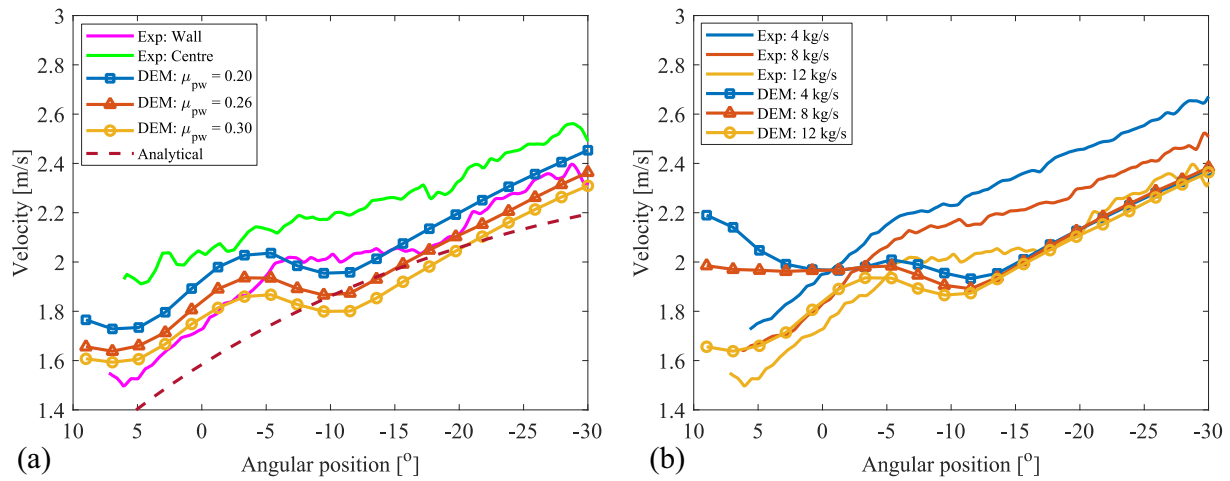


Fig. 12. Transfer hood velocity profile for $V_b = 3 \text{ m} \cdot \text{s}^{-1}$ and (a) $\dot{m} = 12 \text{ kg} \cdot \text{s}^{-1}$ including the effect of μ_{pw} , and (b) for $\mu_{pw} = 0.26$ including the effect of $\dot{m}$.

polycarbonate, $\mu_{pw} = 0.26$ (Section 5.1.3). Slightly lowering the friction to $\mu_{pw} = 0.20$, increased the velocity, and increasing it to $\mu_{pw} = 0.30$ decreased the velocity. The DEM results showed that the material decelerated (Fig. 12a at 10° to 5° and at -5° to -10°), while it accelerated over the other areas of the hood. The measurements against the wall did not show distinct deceleration of the material, but showed the material to have almost constant speed (not accelerating) in these same areas. The DEM model by Ilic et al. [33] also predicted two regions of deceleration, which also reflected in the stream thickness. The analytical model failed to accurately predict the whole velocity profile. This was due to the trajectory model predicting a lower impact position, consequently the impact angle was smaller, which resulted in a lower initial velocity.

The velocity profile for mass flow rates of $4 \text{ kg} \cdot \text{s}^{-1}$, $8 \text{ kg} \cdot \text{s}^{-1}$ and $12 \text{ kg} \cdot \text{s}^{-1}$ at a belt speeds of $3 \text{ m} \cdot \text{s}^{-1}$ is shown in Figure 12b. The results were all taken at the wall of the hood with $\mu_{pw} = 0.26$. DEM predicted that the material velocity at the hood's exit was mass flow rate independent; however, this was not the case in the experiments where the velocity decreased with an increase in mass flow rate. In general, DEM was more accurate for higher mass flow rates.

9. Results: rock box

Due to the position of the rock box, Fig. 7a, the feeding conveyor was set to a speed of only $2 \text{ m} \cdot \text{s}^{-1}$, while the mass flow rate was

set to $4 \text{ kg} \cdot \text{s}^{-1}$, $8 \text{ kg} \cdot \text{s}^{-1}$ and $12 \text{ kg} \cdot \text{s}^{-1}$ respectively. The material characteristics analysed included the material impact velocity V_i , angle α_i , slump/bed angle θ_{bed} , outflow velocity V_o , outflow (discharge) angle α_o and build-up height H_b box, as defined in Fig. 13a. The material slump angle θ_{bed} was defined as the angle which the free surface of the particle bed made with the horizontal, and was measured from the height of the lip to the horizontal centre of the rock box. The material build-up height H_b was defined as the height which the centre of the bed reached against the back wall (see Fig. 13 and Fig. 15). It was found that the discharge angle α_o of the particles varied significantly across the outflow stream, where the particles at the top of the rock box discharged at a relatively shallow angle and the bottom particles at a much steeper angle. Therefore, the discharge angle was evaluated between the upper and lower bounds of the discharge stream, along a line perpendicular to the flow at the centre of the stream.

A summary of the measured flow characteristics is presented in Table 7. These results showed that the mass flow rate had no significant influence on the incoming stream's velocity V_i or angle α_i . However, the mass flow rate significantly influenced the amount of material in the box at any given time, which influenced the slump angle θ_{bed} and material build-up height H_b , both increasing with an increase in mass flow. As a result, the outflow velocity V_o was also influenced by the mass flow rate, due to the increased slump angle.

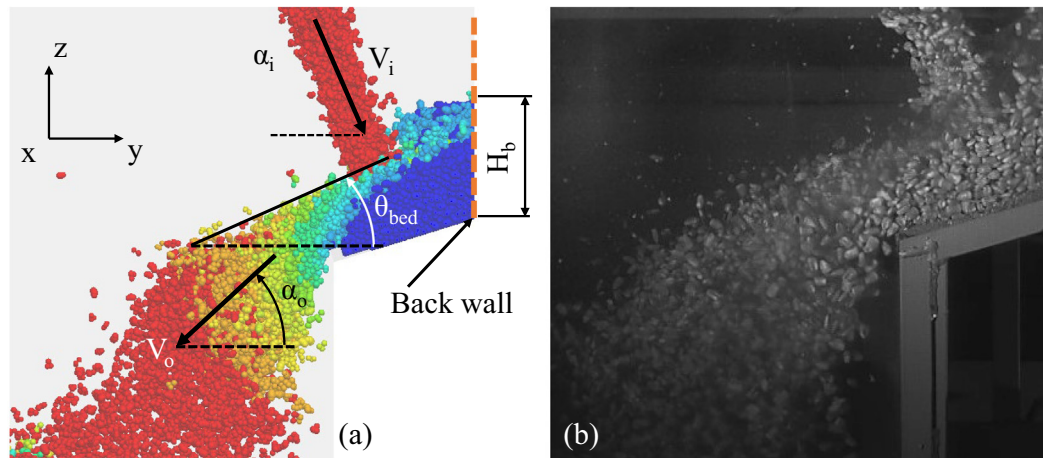


Fig. 13. Rock box material flow from (a) DEM with 3-clump particles and (b) high-speed footage ($V_b = 2 \text{ m/s}$ and $\dot{m} = 12 \text{ kg/s}$).

Table 7Measured impact velocity V_i , impact angle α_i , slump angle θ_{bed} , outflow velocity V_o and build-up height H_b in the rock box.

Mass flow [$\text{kg} \cdot \text{s}^{-1}$]	Impact velocity			Impact angle			Slump angle			Outflow velocity			Build-up		
	V_i [$\text{m} \cdot \text{s}^{-1}$]			α_i [$^\circ$]			θ_{bed} [$^\circ$]			V_o [$\text{m} \cdot \text{s}^{-1}$]			H_b [mm]		
	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max
4	3.88	4.26	5.12	62.4	64.2	66.1	17.9	23.8	29.5	1.18	1.29	1.41	26.6	51.3	87.9
8	3.94	4.26	4.96	62.5	64.6	66.0	19.0	26.9	33.1	1.27	1.35	1.45	45.1	68.7	99.9
12	3.85	4.29	5.08	62.9	63.9	65.1	21.4	31.0	35.2	1.34	1.41	1.50	55.1	75.3	101.9

9.1. The influence of particle friction and damping

Using the 3-clump particles, the particle sliding friction μ_{pp} and contact damping ζ_{pp} were varied between 0.10 and 0.40. The calibrated values of $\mu_{pp} = 0.15$ and $\zeta_{pp} = 0.14$ were also used, Table 8. Similar to the experimental observations, the DEM models also predicted that the mass flow rate had no significant effect on the impact velocity or impact angle. The results compared very well, with the error in impact velocity <4.5%, and the error in impact angle <1.8%. Moreover, these measures were insensitive to changes in μ_{pp} and ζ_{pp} over the ranges investigated.

The trend in the slump angle and material outflow velocity results also correlated well to the experimental observations where it was found that the slump angle and outflow velocity increased for an increase in mass flow rate. However, the results showed that the slump angle was sensitive to the sliding friction, and for $\mu_{pp} = 0.40$, θ_{bed} was significantly overestimated (error of 27% to 42%). However, the θ_{bed} angles predicted by the calibrated value, $\mu_{pp} = 0.15$, agreed very well with the experimental results (error $\leq 6.3\%$). It is interesting to note that for larger friction values ($\mu_{pp} = 0.40$), the slump angle was less sensitive to the mass flow rate, contrary to the experimental results, indicating that these high friction values cannot accurately model the flow mechanism. The material outflow velocity was very well predicted by DEM for all parameter combinations (error $\leq 10.6\%$).

The variation in build-up height H_b over time, was a direct effect of the chevron pattern. This did not significantly affect the impact velocity, impact angle, the slump angle or the outflow velocity. It did however, influence the amount of material “dumped” onto the bed at a given time, and thus the bed height. The measured minimum, average and maximum build-up height H_b are shown in Fig. 14. A box-and-whisker plot was used to present the simulation results and to indicate the

variation in H_b over the simulated time. It is clear that the lower the mass flow rate, the more variation in H_b was found, with the minimum and maximum further apart and a larger interquartile range (IQR). For $\dot{m} = 4 \text{ kg} \cdot \text{s}^{-1}$, the build-up was very well predicted and the IQR fell between the measured upper and lower bounds. The calibrated parameter sets of $\mu_{pp} = 0.15$ and ζ_{pp} equal to 0.14 and 0.40 respectively, were the only combinations where the average measured H_b fell within the IQR. The DEM results for $\dot{m} = 8 \text{ kg} \cdot \text{s}^{-1}$ showed less accurate results where all the IQR's were above the measured average, but still below the maximum. The predicted H_b for $\dot{m} = 12 \text{ kg} \cdot \text{s}^{-1}$ showed the least accurate results and the IQR's fell on or just above the experimental upper bound.

The profile which the material made against the back wall of the rock box (defined in Fig. 13) was evaluated in DEM and compared to measurements, as shown in Fig. 15. Due to the varying material height in the rock box, the average material profile over the simulated time is presented. It was once again noticed that the particle sliding friction had the most significant effect on the material flow while the particle damping had little to no effect. It is clear that using $\mu_{pp} = 0.40$, predicted a significantly higher profile and failed to identify the two lower regions, left and right of the centre. Based on the high-speed footage, it was observed that these lower regions were created by the rolling and sliding of the particles, away from the centre. For lower particle frictions ($\mu_{pp} = 0.10$ to 0.15), the model could accurately predict this phenomenon; however, it failed to accurately predict the depth and size of the lower regions.

The outflow (discharge) angle results are shown in Fig. 16, where the starting position was at the upper bound. It is clear that μ_{pp} and ζ_{pp} only affected the trajectory angle of the upper part of the stream, which was expected since the material in the lower part of the rock box only free-falls once the material flowed over the edge (lip). However, in the upper part of the stream, more particle collisions were

Table 8DEM (3-clump) predicted impact velocity V_i , angle α_i , slump angle θ_{bed} , and outflow velocity V_o in the rock box for various combinations of μ_{pp} and ζ_{pp} .

μ_{pp}	ζ_{pp}	Mass flow $\dot{m}$ [$\text{kg} \cdot \text{s}^{-1}$]	Impact velocity		Impact angle		Slump angle		Outflow velocity	
			V_i [$\text{m} \cdot \text{s}^{-1}$]	Error*	α_i [$^\circ$]	Error*	θ_{bed} [$^\circ$]	Error*	V_o [$\text{m} \cdot \text{s}^{-1}$]	Error*
0.10	0.14	4	4.43	3.99 %	64.7	0.78 %	20.1	-15.55 %	1.20	-6.98 %
		8	4.43	3.99 %	64.8	0.31 %	28.5	5.95 %	1.32	-2.22 %
		12	4.40	2.56 %	64.8	1.41 %	41.4	33.55 %	1.43	1.42 %
0.15	0.10	4	4.43	3.99 %	64.2	0.00 %	21.0	-11.76 %	1.19	-7.75 %
		8	4.44	4.23 %	64.8	0.31 %	27.7	2.97 %	1.33	-1.48 %
		12	4.44	3.50 %	64.9	1.56 %	30.3	-2.26 %	1.49	5.67 %
0.15	0.14	4	4.43	3.99 %	64.6	0.62 %	22.3	-6.30 %	1.16	-10.08 %
		8	4.44	4.23 %	64.9	0.46 %	28.0	4.09 %	1.33	-1.48 %
		12	4.45	3.73 %	65.0	1.72 %	29.5	-4.84 %	1.48	4.96 %
0.15	0.40	4	4.44	4.23 %	64.8	0.93 %	24.3	2.10 %	1.16	-10.08 %
		8	4.45	4.46 %	64.9	0.46 %	27.3	1.49 %	1.28	-5.19 %
		12	4.45	3.73 %	65.0	1.72 %	29.5	-4.84 %	1.43	1.42 %
0.40	0.14	4	4.44	4.23 %	64.5	0.47 %	33.9	42.44 %	1.26	-2.33 %
		8	4.45	4.46 %	64.7	0.15 %	38.3	42.38 %	1.44	6.67 %
		12	4.42	3.03 %	64.7	1.25 %	39.4	27.10 %	1.56	10.64 %

*Error % calculated with respect to the measured value.

Values in red (> 10 %) are considered inaccurate.

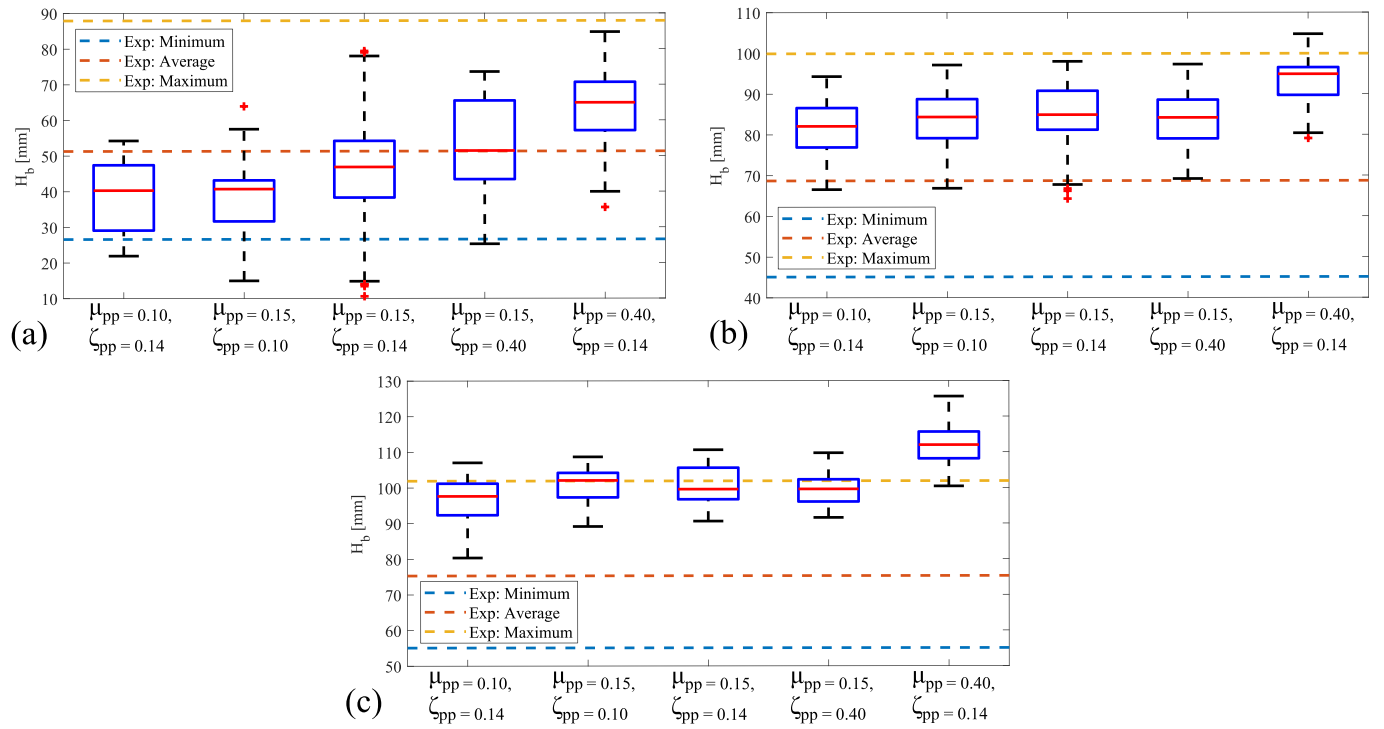


Fig. 14. Predicted material build-up height H_b for various combinations of μ_{pp} and ζ_{pp} at mass flow rates of (a) 4 kg/s, (b) 8 kg/s and (c) 12 kg/s.

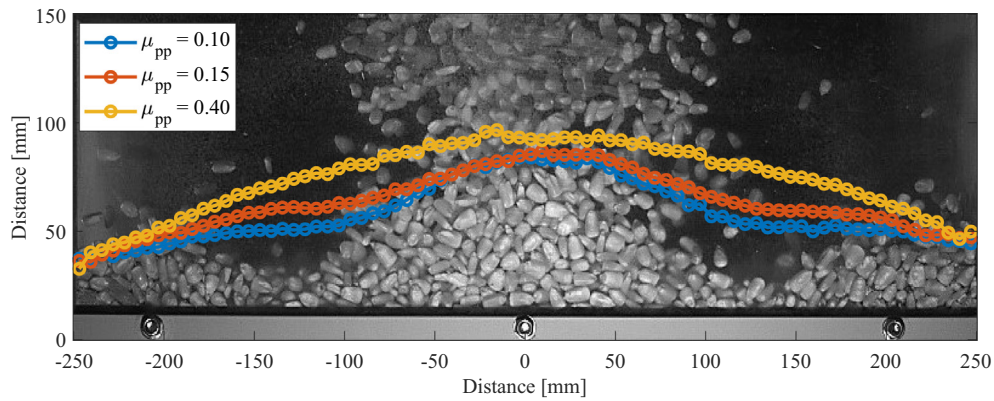


Fig. 15. Influence of μ_{pp} on the material profile against the back wall of the rock box ($\dot{m} = 8$ kg/s and $\zeta_{pp} = 0.14$).

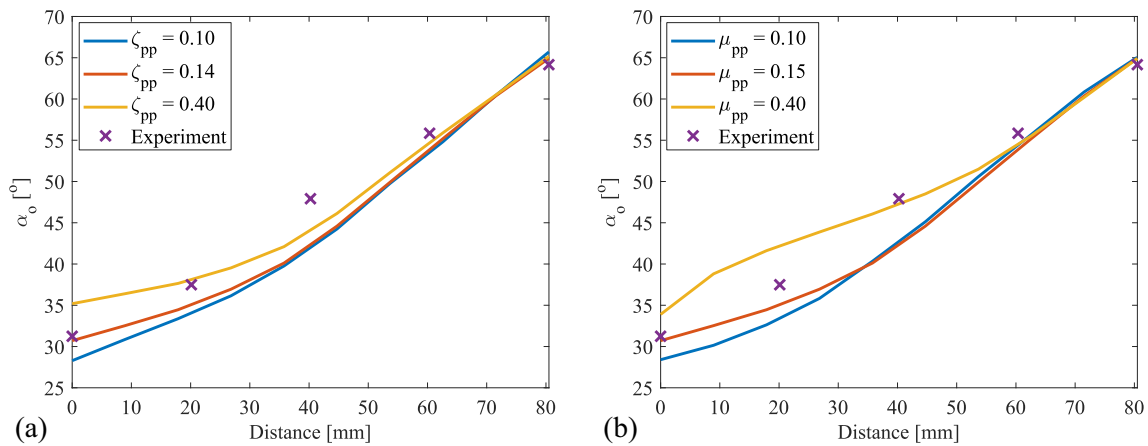


Fig. 16. Influence of (a) ζ_{pp} ($\mu_{pp} = 0.15$) and (b) μ_{pp} ($\zeta_{pp} = 0.14$) on the material outflow angle α_o from the rock box with $\dot{m} = 8$ kg/s.

Table 9Rock box flow characteristics for scaled 3-clump particles ($\dot{m} = 12$ kg/s).

Scale	$\theta_{\text{bed}} [^\circ]$	Slump angle and error		V_o [m/s]	Outflow velocity and error	
		Compared to 1:1 model	Compared to experiment		Compared to 1:1 model	Compared to experiment
1.2	32.6	10.51 %	5.16 %	1.45	-2.03 %	2.84 %
1.4	33.5	19.64 %	8.06 %	1.42	6.77 %	0.71 %
1.6	33.8	14.58 %	9.03 %	1.41	-4.73 %	0 %
1.8	36.1	22.37 %	16.45 %	1.39	-6.08 %	-1.42 %
2.0	35.6	20.68 %	14.84 %	1.38	-6.76 %	-2.13 %

Values in red (> 10 %) are considered inaccurate.

observed which were influenced by the inter-particle parameters. The results further showed that the particle sliding friction had a significant effect on the material flow where a larger discharge angle, i.e. more downwards trajectory, was observed for an increase in μ_{pp} . The particle damping, however, also influenced the discharge angle but to a lesser degree. Furthermore, the DEM results for the sliding friction in the calibrated range, i.e. $\mu_{\text{pp}} = 0.10$ and 0.15 , correlated very well with the experimental results.

9.2. The influence of particle scale

The maximum scaling factor of the 3-clump particles was investigated using the calibrated parameter set of $\mu_{\text{pp}} = 0.15$, $\mu_{\text{rpp}} = 0$, and $\zeta_{\text{pp}} = 0.14$. The particle scale have no significant effect on the material impact velocity V_i and angle α_i . The effect of particle scale on the material slump angle and outflow velocity are presented in Table 9. DEM predicted only a slight decrease in the material outflow velocity with an increase in scaling factor (error <3%). The scaling did, however, had a significant effect on the slump angle where it was observed that θ_{bed} increased with an increase in particle size. Compared to the experimental results, scaling factors ≤ 1.6 were accurate (error <10%), while larger scaling factors significantly decreased the accuracy.

Fig. 17 shows the predicted material build-up height H_b as a function of particle scale. The build-up decreased with an increase in the scaling factor, and the interquartile ranges increased (i.e. more variation in the results). Based on these results, it is concluded that the 3-clumps can be scaled by a maximum factor of 1.8. When using the average particle size of approximately 8.5 mm ($7.5 \text{ mm} \leq D_{\text{ev}} \leq 9.6 \text{ mm}$), the average build-up height was the equivalent of approximately 9 unscaled particles (scale 1:1). Using a scale factor of 1.8, the build-up height was the equivalent of approximately 5 particles, which is the limit to allow for accurate material build-up. The results for the outflow angle, Fig. 18,

showed an increase in α_o (flow more downwards) for the upper material stream with an increase in the scaling factor. However, no difference in the outflow angle was found for the lower material stream.

Although a scale factor of 1.8 accurately predicted the build-up height H_b , accurate modelling of the slump angle, limited the scale factor to 1.6. It is thus concluded, that for this rock box and the corn particles, the maximum scale factor is 1.6 to accurately model all the flow characteristics. However, further investigations should be done to determine the maximum ratio of the particle size to rock box size for accurate predictions in general. With a scale factor of 1, the footprint size of the box ($136 \text{ mm} \times 500 \text{ mm}$, Fig. 7b) was equivalent to approximately 16×59 particles, and for a scaling factor of 1.6 it was 10×37 particles. These results correspond with the general assumption that wall or boundary effects are insignificant if the ratio of the container to particle size is larger than approximately 8 to 10 [67].

9.3. The influence of particle shape

The use of balls to predict the material flow in and from the rock box was investigated. The inter-particle sliding and rolling friction parameters were once again selected from the calibration sets by Coetzee [8] ($\mu_{\text{pp}} = 0.15$, $\mu_{\text{rpp}} = 0.20$; $\mu_{\text{pp}} = 0.20$, $\mu_{\text{rpp}} = 0.125$ and $\mu_{\text{pp}} = 0.25$, $\mu_{\text{rpp}} = 0.125$), and the more accurate parameter set ($\mu_{\text{pp}} = 0.25$, $\mu_{\text{rpp}} = 0.50$), identified in Section 7.7, was also investigated.

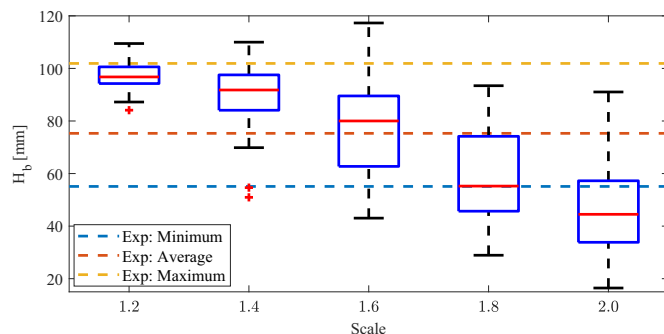


Fig. 17. Material build-up height H_b using scaled 3-clump particles ($\dot{m} = 12$ kg/s)

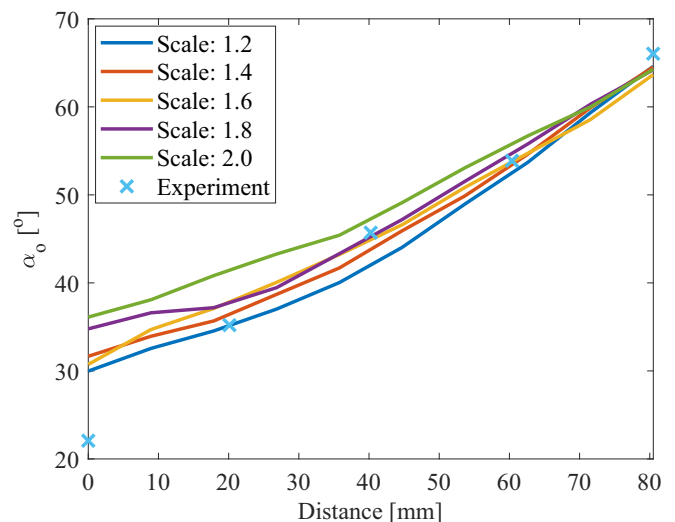
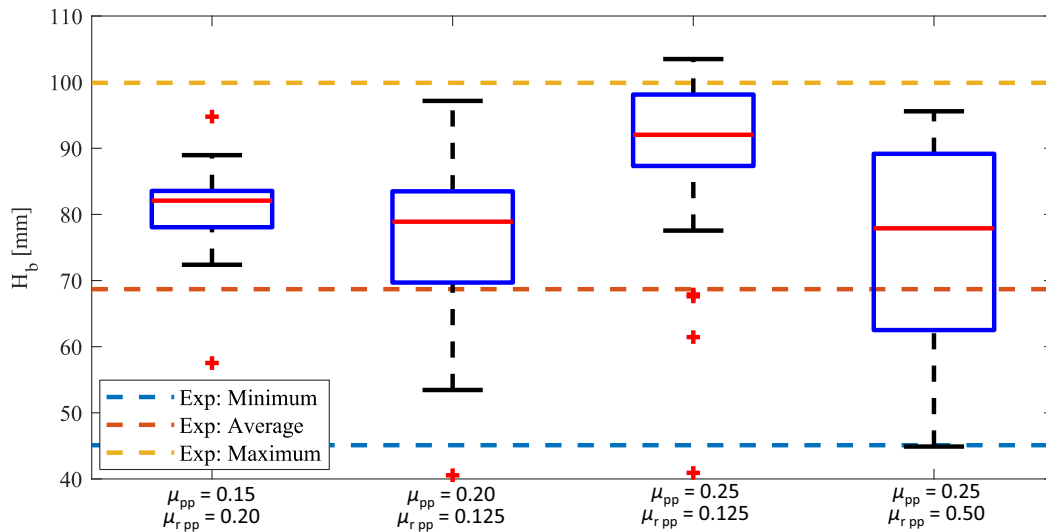


Fig. 18. Material outflow angle α_o using scaled 3-clump particles ($\dot{m} = 12$ kg/s).

Table 10Rock box flow characteristics using balls and combinations of μ_{pp} and μ_{rpp} ($\dot{m} = 8 \text{ kg/s}$).

μ_{pp}	μ_{rpp}	$\theta_{bed} [^\circ]$	Slump angle and error		Outflow velocity and error		
			Compared to 3-clump model	Compared to experiment	$V_o [m \cdot s^{-1}]$	Compared to 3-clump model	Compared to experiment
0.15	0.20	45.1	61.07 %	67.66 %	1.45	9.02 %	7.41 %
0.20	0.125	35.4	26.43 %	31.60 %	1.43	7.52 %	5.93 %
0.25	0.125	28.6	2.14 %	6.32 %	1.42	6.77 %	5.19 %
0.25	0.50	27.1	-3.21 %	0.74 %	1.36	2.26 %	0.74 %

Values in red (> 10 %) are considered inaccurate.

**Fig. 19.** Predicted build-up height H_b using balls ($\dot{m} = 8 \text{ kg/s}$).

The particle shape had no significant effect on the material impact velocity V_i and angle α_i . Table 10 summarises the results for the slump angle θ_{bed} and outflow velocity V_o . The slump angle decreased with an increase in sliding friction μ_{pp} for the balls, while an increase was observed for the 3-clump particles (see Table 8). Furthermore, the parameter combination $\mu_{pp} = 0.25$ and $\mu_{rpp} = 0.125$, calibrated by Coetzee [8], accurately predicted the slump angle and the outflow velocity (error

$\leq 6.3\%$). The parameter set which provided the most accurate results in the impact plate analysis, $\mu_{pp} = 0.25$ and $\mu_{rpp} = 0.50$, was slightly more accurate (error < 1%).

The DEM results for the material build-up height on the rear wall of the rock box is presented in Fig. 19. The parameter combinations proposed by Coetzee [8], showed acceptable results, predicting H_b to be mostly in the upper region of the measured height. However, the parameter combination ($\mu_{pp} = 0.25$, $\mu_{rpp} = 0.50$) identified in the impact plate analysis, showed very good agreement with the experimental results. The minimum and maximum H_b compared very well, and the

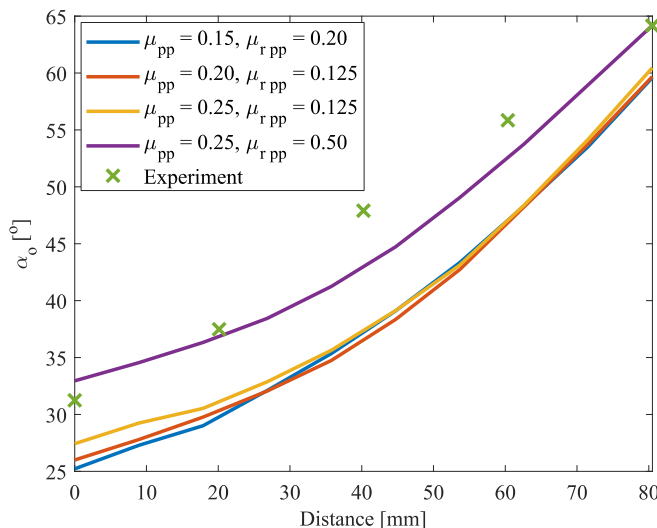
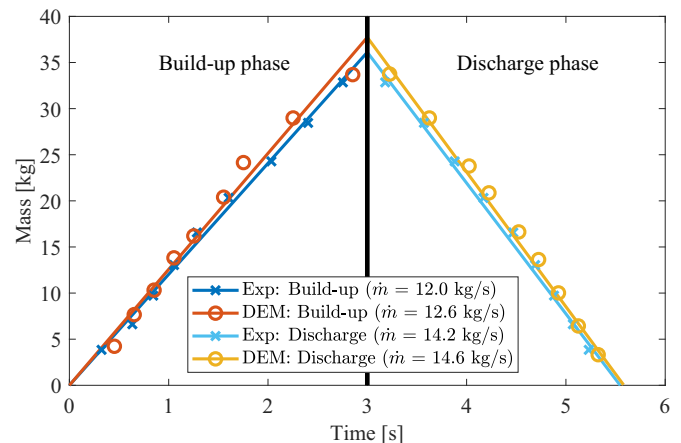
**Fig. 20.** Predicted outflow angle α_o using balls ($\dot{m} = 8 \text{ kg/s}$).**Fig. 21.** Material build-up and discharge mass flow rate results from the transfer chute.

Table 11

Build-up and discharge rates in the chute using scaled 3-clump particles.

Scale	Build-up mass flow and error			Discharge mass flow and error		
	$\dot{m}$ [kg·s ⁻¹]	Error*	R ²	$\dot{m}$ [kg·s ⁻¹]	Error*	R ²
1.0	12.6	5.00 %	0.982	14.6	2.82 %	0.995
1.4	12.9	7.50 %	0.992	14.2	0 %	0.999
1.4**	10.8	-10.00 %	0.956	14.4	1.41 %	0.999
1.6	10.9	-9.17 %	0.983	13.4	-5.63 %	0.998
1.7	9.3	-22.50 %	0.964	12.6	-11.27 %	0.996
1.8	9.1	-24.17 %	0.972	12.3	-13.38 %	0.998
2.2	10.6	-11.67 %	0.981	12.4	-12.68 %	0.999
2.4	9.4	-21.67 %	0.988	11.7	-17.61 %	0.999

*Error % calculated with respect to the measured value.

**Chevron pattern omitted from the receiving conveyor.

Values in red (> 10 %) are considered inaccurate.

interquartile range also spanned the measured range. Comparing these results of the balls to that of the 3-clump particles, Fig. 14b, no significant difference can be observed in terms of the general accuracy.

Fig. 20 shows the simulation results for the material outflow angle α_o . The three parameter sets by Coetzee [8] predicted a much lower outflow angle, showing an approximately 6° offset relative to the experimental results. The angle was, however, accurately predicted by the parameter set identified in the impact plate analysis. This shows that the particle rolling friction had a significant effect on the material bulk behaviour.

10. Results: transfer chute build-up

For the material build-up phase, the mass flow rate was set to $20 \text{ kg} \cdot \text{s}^{-1}$ and the feeding and receiving conveyor speed to $4 \text{ m} \cdot \text{s}^{-1}$ and $0.5 \text{ m} \cdot \text{s}^{-1}$, respectively. Once the material reached a height of 500 mm above the outlet of the chute, the feeding flow was completely switched off and the receiving conveyor's speed was increased to $3 \text{ m} \cdot \text{s}^{-1}$ (discharge phase). The parameter values used are given in Table 2 (those indicated by *) for 3-clump particles with a scale factor of 1.

The predicted and recorded material build-up is shown in Fig. 8a and Fig. 8b, respectively. From visual inspection, the material's free surface was accurately predicted by DEM. The rate of build-up and discharge was calculated from high-speed footage using level markings and the bulk density. Fig. 21 shows the measured and predicted mass inside the chute during both phases. As reference, the material mass was approximately 4 kg at a fill height of 100 mm and 33 kg at 500 mm. A straight line was fitted to the measured data points ($R^2 = 0.995$), and the build-up and discharge rates calculated as $12.0 \text{ kg} \cdot \text{s}^{-1}$ and $14.2 \text{ kg} \cdot \text{s}^{-1}$, respectively.

In the DEM model, the total particle mass inside the chute was summed and recorded (at a frequency of 10 Hz), from which the build-up and discharge rates could be calculated by a fitting a straight line to the data points, Fig. 21. The model accurately predicted the flow rate for both phases with relative errors of 5.0% and 2.8% for the build-up and discharge phases respectively.

The effect of particle scale was investigated, using the 3-clump particles, with the results summarised in Table 11. The results showed a decrease in the flow rate with an increase in the scale factor. Keeping the error $\leq 10\%$, the maximum scaling factor was 1.6, where the error in predicting the build-up and discharge rate was underestimated by 9.2% and 5.6% respectively. With the opening of the chute $80 \text{ mm} \times$

396 mm (Fig. 8), the size can be expressed as 9.4×46.6 unscaled particles, and with the scaling factor of 1.6, this reduces to $\approx 6 \times 29$ particles.

The simulation for a particle scale of 1.4 was repeated without the chevron pattern on the receiving belt. The results showed that the build-up started approximately 1 s later and as a result, the mass flow rate during the build-up phase was also lower. The result, however, still compared relatively well to the measurements (10% error). During the discharge phase, there was no significant difference in omitting the pattern from the conveyor.

11. Conclusion

The accuracy with which DEM can predict the flow of a non-cohesive granular material (corn grains) through a conveyor transfer point was investigated. This included an impact plate, hood, rock box and a chute. The models were validated by comparing the results to experimental measurements, including the use of particle image velocimetry (PIV). The corn grains were modelled as single spherical (ball) and multi-sphere (clump) particles, and the model parameter values were obtained from literature and various experimental tests, followed by a sensitivity study. The conveyor belts had a chevron pattern, which was included in the DEM model to investigate the effects thereof.

It is concluded that DEM could accurately (errors < 10%) predict all the material flow characteristics through a transfer chute, once the parameter values were accurately calibrated. It is important to include the chevron pattern on the belt, if the impact force of the material discharging from the conveyor onto the chute, is to be accurately modelled.

The flow was most sensitive to the inter-particle sliding friction, while the contact damping had less of an effect on the results. The contact stiffness had no significant effect on the results. After each of the shapes (spherical and non-spherical multi-spheres) was individually calibrated, the particle shape had no significant effect on the results. The spherical particles, however, should be modelled using sliding and rolling friction, while the non-spherical particles can be modelled using only sliding friction. Computation time could be reduced by scaling up the particle size (reducing the total number of particles in the model). However, the maximum scaling factor was found to be 9 and 7, without significantly affecting the results in predicting the average impact force and force frequency content, respectively, on a vertical impact plate. The maximum scaling factor for predicting the flow characteristics in a rock box and the rate of material build-up in a chute, was found to be 1.6. Expressed in terms of particle size, the scaling should be such that the smallest dimension of the rock box footprint is at

least 10 particles and the bed height at least 5 particles. The smallest dimension of the chute opening should be at least 6 particles.

The flow behaviour of cohesive (wet) granular materials is different from non-cohesive materials, and more challenging to model accurately. Cohesive materials are also more prone to cause build-up and hang-ups (blockage). The next step should be to calibrate and quantitatively validate DEM models for cohesive materials under dynamic flow conditions, as experienced in a typical transfer chute.

Credit Authorship Contribution Statement

J. Rossow: Conceptualization, Methodology, Supervision, Data curation, Formal analysis, Investigation, Validation, Writing. **C.J. Coetzee:** Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Writing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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